High-dimensional state trajectories reveal transformer inference dynamics

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Abstract

Mechanistic studies localize language-model function in parameters, heads or neurons, but these parts do not show how one inference unfolds. We treat the ordered layer-state trajectory as the empirical process and obtain three bounded findings. First, realized Qwen, Llama and Gemma trajectories are 1.60-2.24 times more chord-aligned than endpoint-preserving shuffles, yet random initializations are more aligned and reordered executions lose native outputs: architecture supplies a propagation scaffold, whereas realized order is functionally consequential. Second, a Pythia-160m audit shows that task function requires compatible input, block and output components; a task-built ΔU coordinate is useful for declared candidate separation but is not superior to a cheap predecision logit-margin baseline. Third, after additive interventions failed, confidence-gated local-transition actuators crossed a declared full-vocabulary boundary in 17/87, 13/87 and 9/87 held-out prompts, with 0/203, 2/203 and 0/203 non-target changes. A matched empirical-subspace direction null did not establish checkpoint-general direction specificity. Ordered high-dimensional states make inference observable and support bounded relation-graph boundary modulation; geometry, forecasting, direction specificity and correctness remain separate limits.

Main text

Large language models produce ordered linguistic and reasoning behaviour despite computations distributed across many parameters and layers[1-4]. Mechanistic interpretability has made those computations experimentally accessible, yet a common description of how an inference unfolds remains elusive[13,14]. The usual explanation is scale. We considered a prior possibility: the difficulty may begin before measurement, in the choice of what is being measured. A mechanism can be implemented by parameters, attention heads, multilayer perceptrons and residual blocks without being most simply expressed as any one of those engineering objects.

Probes establish what information is available in activations, provided that the probe itself is controlled[5,6]. Head and circuit analyses identify local computational motifs[7-9,18]; causal tracing and editing localize consequential associations[10]; vocabulary readbacks expose latent output tendencies[11,19,20]; and sparse-feature methods seek more interpretable activation units[12,21,22]. Each approach answers an important question about an implementing component or an observable. Their natural engineering boundaries, however, do not necessarily define the empirical object on which an inference mechanism is comparable across depth, checkpoints and interventions.

This distinction suggests a methodological ordering: choose a suitable empirical research object before choosing its coordinates. We therefore switched from a catalogue of implementing parts to the ordered inference process and represented it by the sequence of high-dimensional layer states. Layer depth is an ordered evolution coordinate, not physical time.

Comparable methodological choices arise in other high-dimensional systems. Physical dynamics may require informative state or reaction coordinates, whereas systems neuroscience can reveal population trajectories that are not apparent from single-unit descriptions[23-25]. We use these as sources of testable questions, not as evidence that transformers share physical or neural mechanisms.

Existing representation-comparison methods ask related but different questions. SVCCA and related similarity measures compare representation sets through chosen invariances and geometries[15,16]; representation-manifold paths compare how networks transform data across depth[17]. Our unit is one prompt-conditioned inference trajectory. We retain its realized block order, perturb that order under a fixed-endpoint null and intervene on its local transitions. Vocabulary projections and compressed views can therefore be audited as coordinates of this process rather than assumed to be the process itself.

The results resolve into three linked but non-interchangeable conclusions. First, direct hidden states provide stronger mechanism coordinates than vocabulary readbacks in the tested audits, but the compact candidate-separation coordinate ΔU is not required to outperform a recent logit-margin baseline. Second, layer order carries a reproducible geometric signature while random initialization, execution reordering and component swaps separate architectural propagation, trained coordinate compatibility and functional execution. Third, a confidence-gated local-transition actuator crosses a declared emitted-token boundary under a relation-graph contract, whereas its checkpoint-general direction specificity and cross-task output control remain unconfirmed. The sections that follow test these three conclusions in turn: coordinate audit and bounded registration, architecture/training/order separation, and perturbational boundary control. The resulting chain establishes an observable process and bounded control of a declared emitted-token boundary within one relation-graph task family across all three checkpoints. It does not establish a complete predictive theory, exact geodesic law or answer-correctness control.

Layer-state trajectories expose an ordered process

For checkpoint m, prompt p and layer l, let $h_{m,p,l} \in \mathbb{R}^{d_m}$ be the last-token hidden state and

$$T_{m,p} = (h_{m,p,1}, \ldots, h_{m,p,L_m})$$

the ordered inference trajectory. The definition preserves two features that disappear when a model is reduced to a parts list: the state visited at each depth and the order in which those states are visited. We used parameters and modules to extract the trajectory, but treated the trajectory itself as the empirical process object (Fig. 1a). Its high dimensionality was retained in the analyses; the low-dimensional orientation view in Extended Data Fig. 1 is not evidentiary.

The object switch changes the question from which component contains a feature to how a prompt-conditioned state evolves through the components that implement it. Adjacent hidden-state cosine distances changed non-uniformly with depth, and the normalized participation ratio of each update showed that its energy remained distributed across many hidden dimensions (Fig. 1b,c). These measurements use native normalized states rather than a vocabulary projection. They establish an ordered, distributed process object, not continuous physical time or checkpoint-identical coordinates.

Each prompt presented a relation graph and required one of two one-word labels. ”Ava belongs to Bela” and ”Bela maps to Red” imply Red for Ava. Red is the clean candidate from the unperturbed chain; an added statement introduces Blue as the conflict candidate. Before fitting, prompt metadata were collapsed into three analysis labels: stable prompts leave the original route decisive, competition prompts support both labels, and closure prompts update or override the route, allowing the conflict candidate to become rule-consistent. These are predefined prompt-condition classes, not dynamical states discovered from the trajectories. Clean and conflict mark provenance, not correctness. Neither ΔU nor model outputs entered these labels; the coordinate-audit and intervention panels used separately fixed mappings (Supplementary Methods Table 1), and the testbed does not represent open-ended reasoning.

We next audited the coordinates used to observe that process. In a five-fold relation-graph-grouped test with coordinate-audit class labels fixed by prompt condition, hidden decision state, hidden precursor state and direct hidden flow increased macro F1 by 0.448, 0.448 and 0.444 in Qwen; 0.520, 0.520 and 0.485 in Llama; and 0.240, 0.248 and 0.244 in Gemma. Learned-output-matrix readbacks increased it by 0.407, 0.268 and 0.217, respectively (Fig. 2a). The advantage was checkpoint-dependent but did not require ΔU. Readbacks remain informative output-facing diagnostics, but their geometry depends on the projection matrix and selection rule. Real, row-permuted and Gaussian projection controls did not give the learned readback the preservation advantage expected of a primary process coordinate (Fig. 2b).

The decision-window clean-relative margin profile is retained separately as a construction audit (Extended Data Fig. 2). Because ΔU is the first principal component of the same matrix, that profile is not independent evidence. Removing this circular comparison leaves the group-held-out mechanism-label result and projection controls as the basis of the coordinate audit.

A frozen predecision comparison sets a further boundary on ΔU. At the same layer cutoffs and group-held-out partitions, ΔU_{pre} did not improve mechanism-label macro F1 over matched recent output-logit margins in Qwen, Llama or Gemma (differences -0.064, -0.165 and -0.045; family-adjusted $q \geq 0.998$). Complete predecision margin histories and cutoff hidden states did improve on this recent readback (all $q = 1.50 \times 10^{-5}$). Because no unordered-history comparator was included, this is evidence for retained predecision information, not an order-only gain. ΔU is therefore a compact task construction, not a necessary or superior coordinate (Supplementary Methods Table 9).

Exploratory registration bounds candidate-path competition

Absolute layer number is a poor default correspondence across checkpoints with different depths and internal schedules. We therefore used functional registration as an exploratory alternative to layer-number matching. Sliding windows within the first 45% of depth were scored for clean-relative preservation, prompt-condition-class information and coupling to a later dominance coordinate. The selected intervals differed across checkpoints: 15-35% relative depth for Qwen, 0-25% for Llama and layers 0-6 for Gemma (Fig. 2c,d). Their scan scores supplied model-specific analysis windows; they did not establish homologous locations or shared coordinates.

These windows are exploratory and data-selected. The same scan supplied both candidate windows and the score used to select them. Grouped cross-validation protected the within-window classification and correlation estimates from group overlap, but selection was not nested. We subsequently froze the selected windows and tested 24 new graph groups with disjoint entities, relation phrases and candidate labels. Prompt-condition-class information transferred in all three checkpoints (macro F1 0.739, 0.662 and 0.649; each above its label-permutation null), whereas downstream coupling exceeded its target-permutation null in Qwen and Gemma but not Llama. A predeclared topology-separation test reversed sign in all three checkpoints, and only Gemma retained a top-quartile frozen-window rank (Supplementary Methods Table 2). The three-checkpoint confirmation gate therefore failed. Together with weak pairwise direction isomorphism (Extended Data Fig. 2), this leaves the windows as model-specific exploratory indexing choices, not confirmed functional phases.

Within the controlled tasks, the clean-relative signed margin $dR_{p,l}$ compared support for the task-defined clean and conflict candidates relative to the clean condition in the same graph group. Principal-component analysis reduced the decision-window matrix $D = [dR_{p,l}]$, and its oriented first-component score defined ΔU (Fig. 3a). This task-constructed exploratory diagnostic describes separation between declared candidate supports; it is not an independently discovered latent variable. It can be computed from a trajectory, audited under information-removal controls and used as a held-out intervention endpoint. Reaction-coordinate compression motivated the construction, but ΔU is not an energy, physical potential or universal order parameter.

The correlations between ΔU and the related dominance readout (0.9990, 0.9989 and 0.9997) are construction checks, not independent evidence. Predecision recomputation excluded decision-layer and endpoint features; candidate-margin-excluded flow removed the direct margin; topology-window and auxiliary-band controls changed the interval (Fig. 3b-d). Mechanism macro F1 ranged from 0.771 to 0.954, and PC1 explained 0.832-0.967 of decision-profile variance. These tests ask whether accompanying trajectory signatures survive removal or displacement of construction information; they do not establish out-of-task generalization. Final correctness did not enter the construction. We therefore treat ΔU as bounded in inferential role, not numerical range.

Layer order separates geometric scaffold from function

A useful process object should do more than separate conditions. It should also make observed updates compactly describable. We compared state-only features, previous-transition history features, state plus a realized continuous transition coordinate O_l^{cont}, and a bilinear state-transition model (Fig. 4a). Previous-transition history features summarize reversals, changes in relative rank and backtracking before the current transition. They are distinct from the decision-window clean-relative margin profile.

The realized coordinate O_l^{cont} comprises continuous components of the observed adjacent transition. Adding it raised macro R^2 to 0.980, and the bilinear state-transition variant reached 0.993. This is descriptive closure because the realized coordinate contains target-transition information. In a separate pre-transition test, the state-only model reached 0.822, the observed-coordinate model reached 0.979 and the best estimated-coordinate bilinear model reached 0.862, retaining 25.1% of the observed-coordinate gain (Extended Data Fig. 4). Across 768 held-out prompt groups, its macro-R^2 increment over state only was 0.0394 (group-bootstrap 95% interval, 0.0368-0.0426). History and context predicted the coordinate itself with macro R^2 values of 0.41-0.61. Thus target-excluded estimates carry finite prospective information, but O_l^{cont} still compactly factorizes the realized update far better than the estimated coordinate and does not yet supply an equivalent forward law.

We therefore tested a narrower prospective question without using O_l^{cont}: can an ordered partial trajectory forecast the exact final clean-minus-conflict output margin for unseen graph groups? Ridge models used only the current and earlier decision-window margins, with 67 graph groups for fitting and 29 held out. At the earliest cutoff whose held-out R^2 exceeded the 95th percentile of 50 independent within-prompt order nulls, R^2 was 0.574, 0.393 and 0.714 for Qwen, Llama and Gemma; boundary AUROC was 0.857, 0.837 and 0.984 (Extended Data Fig. 6c). This is a bounded forecast of one declared candidate margin. A stronger cross-task test asked whether ordered-prefix summaries added held-out information beyond the current state: 6 of 18 initial conditions passed, all in addition; lexical prompts passed 0/9 and an independent mixed-arithmetic replication passed 0/9 (Extended Data Fig. 7c). Thus the tested histories do not establish a general predictive law; the current high-dimensional state may already summarize much of the usable path history.

We next tested finite-path directional organization in the native hidden-state trajectories. States were normalized to unit length; adjacent steps were compared with the endpoint chord, and detour and turning curvature were calculated. Each of 50 checkpoint-level null repeats preserved both endpoints and all intermediate states but applied one shared random layer permutation to the 96-prompt controlled subset. Observed order was 1.60-2.24 times more chord-aligned than the null mean across Qwen, Llama and Gemma, corresponding to 5.11-5.63 null standard deviations; detour and turning curvature were also reduced (Fig. 4b,c).

This order effect recurred in controlled lexical and addition tasks and a 96-item four-choice ARC-Challenge audit without binary reduction[26]: real order exceeded all 50 shuffles in every task-checkpoint combination (null-standardized effects, 3.21-7.84; Extended Data Fig. 6a,b). Yet three random models from each checkpoint configuration showed still larger alignment gains, and stronger relational-continuity nulls passed in all 18 architecture-task-seed conditions (Extended Data Fig. 7a). Chord-directed order therefore exposes an architecture-compatible propagation scaffold, not learned inference by itself.

We then froze 24 additional graph groups, their ten prompt conditions and the endpoint-preserving shuffle protocol before extraction. On the identical 240-prompt manifest, all-layer native trajectories from Qwen-1.5B and the larger, later-generation Gemma-3-12B-it-QAT-Q4_0 checkpoint exceeded every one of 50 shared order shuffles in chord alignment (5.63 and 3.39 null standard deviations; two-test adjusted $q = 0.0392$; Supplementary Methods Table 11). This confirms the order-null geometry in these two checkpoints and new graph instances, not a pure scaling law, a quantization-invariant measurement or a functional-control result.

Geometry was not sufficient for function. In Qwen, reversed and fixed-permutation executions retained positive chord contrasts but had zero full-vocabulary top-1 agreement with native execution, candidate agreement of 0.479 and mean Jensen-Shannon divergence of 0.692 (Extended Data Fig. 7b). In Llama, direction-only trajectories passed 0/7 task gates whereas norm-only fixed-direction trajectories passed 7/7, and radial rescaling changed the metric monotonically in all seven tests (Extended Data Fig. 7d). Native order is functionally consequential, but raw alignment is scale-sensitive and cannot stand in for learned computation.

To isolate what training adds, we swapped embeddings (E), blocks (B), final normalization (N) and output head (W) between initialized and trained Pythia-160m checkpoints. On 53 held-out lexical cloze prompts, accuracy rose from 0.472 to 0.849 when all components were trained; no isolated trained component exceeded 0.604, and the superadditive margin excess was 2.621 (95% confidence interval, 1.252-3.987; Extended Data Fig. 8a,b). Across seven checkpoints, model-native candidate-coordinate order gain emerged from step 1,000 and correlated with accuracy (Spearman $\rho = 0.75$). A coordinate fitted at the final checkpoint transferred poorly at steps 0 and 128, reached macro F1 0.701 at step 1,000 and 1.0 by step 16,000. The within-checkpoint category control was already perfect at initialization and was rejected as evidence of learning (Extended Data Fig. 8c,d). Training therefore organizes compatible input, transition and output coordinates rather than placing task function in any isolated component.

These metrics describe directional efficiency under a specified cosine geometry and shuffle null; they neither learn a Riemannian metric nor test a geodesic equation. Residual-only, parallel and perpendicular proxies remained complementary (Fig. 4d,e), excluding a one-dimensional collapse. We therefore use chord-directed as the empirical term and withdraw geodesic-biased as a result label; vocabulary-readback-centre trajectories remain auxiliary diagnostics (Extended Data Fig. 3c).

An actuator switch enables trajectory and output-boundary control

Observation and compact description do not by themselves establish that the trajectory account identifies an actionable process. We therefore treated intervention as a perturbational test. The first additive state-direction interventions did not produce robust mechanism-specific control. A subsequent additive flow-direction intervention also failed to produce the required selectivity (Fig. 5a). These negative results ruled out the idea that any correlated direction would serve as an actuator.

The failure prompted a change of intervention object. Adapting the control-theoretic separation between state estimation and actuator choice, we fitted a local transition operator conditioned on the predicted intervention-class label. A policy first interpolated between the realized next-layer state and the operator proposal, then applied a scaled precursor direction. The classifier distinguished predefined stable, competition and closure prompt classes; the policy selected the two weights for held-out graph groups. These classes organize the perturbation test and are not inferred dynamical phases. Final answer correctness was not a classifier target.

The resulting policy increased internal trajectory-dominance specificity over matched fixed combinations in each checkpoint (Fig. 5b). In the 50-shuffle replication, the real policy exceeded the joint mechanism-and-policy shuffle null with z scores of 4.38 for Qwen, 4.23 for Llama and 4.02 for Gemma (Fig. 5c). Because the learned actions were applied to held-out graph groups before the endpoint was measured, these contrasts provide perturbational evidence for selective causal control of the measured internal answer-formation trajectory. Mechanism macro F1 ranged from 0.827 to 0.927, whereas policy macro F1 ranged from 0.534 to 0.708, which bounds the precision of actuator assignment (Fig. 5d).

The original action budget reached the internal endpoint but not every candidate boundary. Clean-candidate selection changed by 0.0, 9.2 and 6.9 percentage points in Qwen, Llama and Gemma (Extended Data Fig. 5). Qwen nevertheless moved 86 of 87 prompts towards clean and shifted the mean margin from -5.03 to -2.54; the maximum remained -0.56. Cross-fitted ΔU-to-margin maps reached pooled $R^2 = 0.850$ and separated 15 original-budget crossings with AUROC 0.976 (Methods; Source Data). Qwen's larger deficit and 94% saturation at the maximum 0.3/0.3 action identified an exhausted budget rather than absent output leverage.

We therefore expanded the bounded action library while fitting all coordinates, transition models and boundary objectives on training graphs only. The largest operator and precursor weights were 1.0 and 1.2, versus the original 0.3/0.3 budget; this was a discrete search, not a dose-response study. Under one predeclared confidence-and-entropy gate, full-vocabulary top-1 changed from conflict to clean in 17/87, 13/87 and 9/87 held-out closure prompts in Qwen, Llama and Gemma, with 0/203, 2/203 and 0/203 non-closure changes (Fig. 5e). Among gate-open prompts that were conflict top-1 at baseline, 17/84 Qwen, 13/40 Llama and 9/17 Gemma prompts crossed. Qwen non-crossings usually moved towards clean from more distant baselines; some Llama and Gemma pairwise crossings remained behind a third token (Supplementary Methods Table 5). Fifty safety-matched reassignments preserved gate coverage and condition-by-layer action multisets; observed counts did not exceed their 95th percentiles ($P =$

$0.804, 0.922, 0.255$; Extended Data Fig. 6e,f). In a separately frozen component ablation, retaining the gated operator dose while setting the precursor weight to zero produced 17, 14 and 8 strict crossings; retaining only the precursor produced 0, 1 and 0. A uniform maximum action on the same gate-open sets produced 22, 15 and 10 crossings, with 0, 2 and 1 non-closure changes. Thus the local transition operator carries the principal output-boundary leverage at this operating point, whereas the precursor alone is insufficient and fine-grained dose assignment remains unsupported. Under the provenance definitions above, crossing controls the emitted token but does not imply an accuracy improvement.

A stricter direction test replaced only the perturbation orientation. It sampled 50 directions per checkpoint from the training-group empirical update subspace while preserving every operator and precursor component norm, layer, gate, action schedule and endpoint. Structured specificity did not exceed the 95th percentile in Qwen or Llama; Gemma exceeded its unadjusted null, but not the three-checkpoint family-adjusted threshold ($P = 1/51$, $q = 0.0588$; Supplementary Methods Table 10). The boundary crossings therefore support the declared gate/executor intervention within this task family, but do not establish checkpoint-general direction specificity.

We then tested transfer under a frozen mixed-arithmetic protocol. Sixty-four problems supplied six matched prompt conditions; 44 problem groups were used for task-specific fitting and 20 were held out. Model-specific intervention windows, the action library, regularization and gate thresholds were inherited without retuning. At a raw-prompt candidate-token boundary, the guarded actuator produced 2 strict crossings among 45 baseline-eligible held-out target prompts and changed 1 of 240 non-target top-1 outputs (Extended Data Fig. 9). Pooled selectivity exceeded 50 address-reassignment repeats that preserved each model's action-dose multiset (one-sided empirical $P = 0.0196$). Only Qwen, however, passed the predeclared model-level gate requiring at least five eligible prompts, a positive median candidate-margin shift and at least one strict crossing; Gemma had only four eligible prompts. This is partial actuator transfer, not confirmatory cross-task output control.

Discussion

Across the tested checkpoints, realized transitions formed a measurable internal process and declared gated perturbations crossed an emitted-token boundary. The new controls separate three levels that raw geometry had conflated: architecture supports relational propagation, training organizes compatible task coordinates and native layer order executes functionally consequential transitions. The ordered trajectory supplies the common empirical object on which these levels can be compared and perturbed. This is process-level perturbational evidence with bounded behavioural leverage, not a universal dynamical law; the new direction null also excludes a checkpoint-general orientation-specificity claim.

The measurements draw on restricted mathematical and physical ideas. Depth indexes a discrete trajectory; checkpoint comparison resembles functional registration; ΔU adopts reaction-coordinate compression without claiming diffusion optimality, a committor or a slow variable. Transition closure tests compact representation of an observed update, whereas chord alignment, detour and turning angle test finite-path directional efficiency. The actuator separates state estimation from local action, and its gate abstains under uncertainty. These are sources for executable tests, not claims of physical identity.

The cross-disciplinary implication is a transferable audit, not a mechanism analogy. Given measurable states and transitions, ask whether a process object supports a compressive coordinate, rejects an endpoint-only explanation under an order-sensitive null, and responds at a predeclared boundary to targeted perturbation. Artificial systems make these tests unusually direct. A systems-neuroscience analogue would compare population trajectories with neuron-wise or static summaries, then perturb an inferred transition against a behavioural boundary. Neural trajectories evolve in physical time, whereas transformer depth is a computational index; appropriate states, timescales and interventions would differ. Our data neither map layers onto neural time nor imply shared mechanisms[23-25].

This view complements component-level interpretability. Circuits can identify which parts realize a local computation; probes and sparse features can identify information available at a state; readbacks can expose output-facing tendencies; and causal interventions can localize consequential variables. A trajectory-level object asks a different question: how do those local objects participate in an ordered process? The answer can then guide where component analyses and interventions should be placed.

Rival explanations remain. Proxy ranking is task- and classifier-dependent. Geometry recurred in lexical, addition and ARC tasks, but task coordinates transferred less consistently and the relation-graph ΔU axis was not reused. Prospectively tested functional windows retained class information but failed the joint three-checkpoint recurrence gate, so their locations remain exploratory. The relation-graph margin forecast did not generalize to an incremental path-memory law; target-excluded transition estimates retain only one quarter of the descriptive O_l^{cont} gain. The training audit used one small Pythia checkpoint and one lexical task, and component swaps are artificial states rather than natural training trajectories. Stronger random-state geometry, radial sensitivity and reordered executions jointly show that geometry alone is neither learned reasoning nor functional fidelity. Failure-mode decomposition showed that Qwen non-crossings were dominated by insufficient displacement from more negative starting margins, whereas Llama and Gemma also encountered third-token competition. Gated control changed 2 of 203 non-closure outputs in Llama. Component ablation located the principal leverage in the local transition operator, yet matched reassignment and uniform-action controls supported no fine-grained dose specificity. The mixed-arithmetic extension showed pooled selectivity but failed its two-model confirmation gate. These are not safety guarantees, candidate crossing is not correctness, and confirmatory output control remains bounded to the relation-graph family.

A broader LLM dynamics landscape may include architecture supplying a propagation scaffold, training organizing a task-compatible representational substrate, prompts conditioning initial states and available directions, and transformer layers realizing local transitions. The Pythia audit provides a bounded training-time bridge: fixed task coordinates emerge and function is distributed across compatible components. It does not map a complete training-shaped semantic geometry or establish how prompt-conditioned directions are formed. Those questions require new experiments rather than extrapolation from the present trajectory data.

The transferable lesson is to choose the empirical research object before its coordinates. This may guide other distributed systems whose parts are accessible but process organization remains obscure, although no transfer is claimed here. The broader question is when a chosen object makes distributed computation observable, computable and controllable. We establish one bounded artificial case; extending it towards a general theory of cognition requires concepts and experiments beyond this paper.

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Methods

Checkpoints, prompts and hidden-state extraction

We analysed representative checkpoints from the Qwen, Llama and Gemma families: Qwen2.5-1.5B-Instruct, Llama-3.2-1B-Instruct and Gemma-2-2B-it (exact revisions in Extended Data Table 1). The principal coordinate-audit dataset contained 960 prompt-condition traces per checkpoint. Analyses requiring identical cross-checkpoint conditions and repeated geometry nulls used a 96-prompt controlled subset. Eight complete graph groups were sampled with seed 20260610, and all 12 conditions in each group were retained. This preserved complete within-group panels and equal condition counts while keeping 50 matched shuffles computationally auditable. The subset was not the first 96 rows.

For each prompt, we extracted the last non-padding-token state from each transformer block output and excluded the embedding layer. Qwen, Llama and Gemma contributed 28, 16 and 26 layer states with hidden dimensions 1,536, 2,048 and 2,304, respectively. An inference pass was represented as $T_{m,p} = (h_{m,p,1}, \ldots, h_{m,p,L_m})$. Layer depth is an ordered evolution coordinate. It is not interpreted as physical time.

Readback diagnostics and projection controls

The relation-graph task defined two candidate labels for every prompt. The clean candidate followed the unperturbed relation chain, whereas the conflict candidate was introduced by a distractor, competing statement, update or exception. In the 12-condition coordinate-audit panel, the stable class contained clean, rename, irrelevant, paraphrase and weak-distractor rows, with the last stored as stable-shift; the competition class contained balanced competition, direct conflict, equal evidence and source claim, with the last stored as source-ambiguity; and the closure class contained closure update, closure override and exception override. Labels came from controlled prompt metadata, not from ΔU, a fitted classifier or the generated answer. The intervention panel used its own predeclared mapping (Supplementary Methods Table 1). Relation graph was the grouping unit for every split.

Vocabulary-facing margins were calculated from the output matrix W_m. For the task-defined clean candidate token c_p and conflict candidate token e_p,

$$R_{p,l} = h_{p,l}^{\top}(W_{c_p} - W_{e_p}).$$

Vocabulary-neighbourhood features summarized the top-ranked vocabulary items under this projection at each layer. Projection-selection geometry controls replaced the real output matrix by either a row-permuted matrix or a seeded Gaussian matrix while retaining the hidden states and analysis pipeline. These analyses test coordinate dependence of readback geometry. They do not test whether readbacks contain any task information and do not make a readback the primary process object.

The decision-window clean-relative margin profile is defined as the vector of clean-relative margins over decision layers. Because ΔU is derived from the same matrix, this quantity is a construction-aligned reference. The Source Data manifest maps repository fields to these reader-facing definitions.

The independent coordinate audit augmented the same predecision base state with one proxy family at a time. Proxy dimensionality was capped at 32; when reduction was required, principal-component analysis was fitted only in each training fold and applied to its held-out fold. Five-fold group cross-validation treated each relation graph as indivisible, so no prompts from one graph occurred in both training and test data and each graph was held out once. Mechanism macro F1 was the unweighted mean of class-specific F1 scores; its reported gain was the augmented-model score minus the base-state score. These labels were fixed before model fitting and did not use ΔU or generated answers.

Exploratory functional-window scan and prospective boundary test

Cross-checkpoint windows were selected by a sliding scan over the first 45% of model depth. Candidate window lengths were 2, 3, 4, 5, 6 and 8 layers with a one-layer step; readback neighbourhood sizes $K = 100$ and $K = 500$ were evaluated. For each window, a preservation score was calculated as one minus the mean clean-relative distance for relation-preserving conditions. The source code stored this quantity under a topology field, but it is not itself a topology-separation test. Prompt-condition-class information was measured by group-aware cross-validated macro F1, and downstream coupling by a group-aware cross-validated correlation with ΔU. The exploratory objective was

$$S = 0.5 S_{\text{preservation}} + 0.8 \max(S_{\text{class}}, 0) + 0.8 \max(S_{\text{downstream}}, 0).$$

The maximum-scoring interval was retained for each checkpoint. Selection and evaluation used the same scan. Group-aware folds prevent graph-group overlap within the component estimates but do not correct selection optimism because window selection was not nested.

For the prospective boundary test, the selected window and K were frozen before extraction of 24 new graph groups. The confirmation set retained the 12 condition templates but used six new entity triples, four new relation-phrase pairs and 12 candidate labels disjoint from discovery. The discovery rows alone fitted the decision-profile PCA, feature scaling, three-class logistic classifier and ridge model. Confirmation labels were used only for scoring. Two thousand graph bootstraps and 999 discovery-label or within-graph target permutations supplied intervals and nulls. Per checkpoint, recurrence required a positive topology-separation interval, macro F1 at least 0.55 and above the label-null 95th percentile, and positive downstream correlation above its target-null 95th percentile. Frozen-window location was tested separately against the unchanged candidate pool, with top quartile declared in advance. No checkpoint passed the joint role gate because topology separation was negative in all three; Qwen and Gemma passed the two predictive endpoints, whereas Llama passed class information only. Frozen-window ranks were 58/112, 38/40 and 5/100. We therefore retain the intervals as exploratory indexing choices and do not interpret them as confirmed or universal phases.

Construction and boundary of ΔU

Within each graph group, the clean condition supplied a layerwise baseline $R_{p_0,l}$. For every condition p, the clean-relative margin was

$$dR_{p,l} = R_{p,l} - R_{p_0,l}.$$

If a group lacked a clean row, the implementation used the group mean as the baseline. Let $\mathcal{L}_{\text{dec}}$ be the mapped decision-layer set and

$$D = [dR_{p,l}]_{p,\, l \in \mathcal{L}_{\text{dec}}}\,.$$

The task metadata specify a clean candidate and a competing conflict candidate, and these candidate identities enter R and D. The model's final generated answer and its correctness label do not. Principal-component analysis was fitted to D, and ΔU_p was the score on PC1. Its sign was oriented, where possible, so that stable or stable-shift rows had larger mean scores than closure rows. The decision-window clean-relative margin profile is the row $D_{p,:}$ itself.

Predecision controls rebuilt the predictors after excluding decision layers and endpoint-derived features. Candidate-margin-excluded controls used neighbourhood-flow variables that did not include the direct clean-versus-conflict margin; topology and auxiliary-band controls changed the analysed layer interval. The correlation of ΔU with a dominance readout built from related decision-window information is treated only as construction sanity. ΔU is a bounded candidate-path separation observable, not an energy, a final-answer label or independent proof of mechanism.

State-transition closure

Closure analyses compared feature families for describing the observed next-layer state structure. State-only features contained the selected current-state variables. Previous-transition history features encoded prior clean-relative margin changes, prior rank-gap changes, reversals in each signal and two backtracking indicators. The realized continuous transition coordinate O_l^{cont} comprised the first ten principal components of the observed transition representation. State plus O_l^{cont} and bilinear models used

$$\hat{y}_{p,l+1} = f(z_{p,l}, O_l^{\text{cont}})$$

and

$$\hat{y}_{p,l+1} = f(z_{p,l}, O_l^{\text{cont}}, z_{p,l} \otimes O_l^{\text{cont}}),$$

respectively. Closure was summarized by macro R^2, the unweighted mean of target-wise R^2 values, together with variance-weighted R^2, RMSE and MAE where available. When O_l^{cont} was calculated from adjacent states, the analysis is descriptive factorization because target-transition information enters the coordinate.

For the predictive boundary test, $\hat{O}_l^{\text{cont}}$ was estimated under graph-grouped cross-validation from current-state features, previous-transition history and, in the richest model, layer and prompt-condition context.

The target layer and realized transition were excluded from these predictors. Within each outer fold, independently predicted training- and test-fold coordinates were combined with current-state variables and their bilinear products. All preprocessing and both stages were fitted on training groups only. The corrected branch produced complete out-of-fold predictions for all four feature sets; unaffected state-only and realized-coordinate branches were numerically invariant to the correction. Supplementary Methods Table 1 maps the information boundary for each analysis family.

Prospective final-margin forecasting

We separately forecast the exact final clean-minus-conflict output-logit margin from partial decision-window histories. The frozen relation-graph split contained 67 training groups (670 prompts) and 29 held-out groups (290 prompts). At cutoff k, the ordered-history ridge model used $R_{p,l}$ only for decision layers $l \leq k$; no later layer state, realized transition, final generated token or correctness label entered its predictors. Ridge penalty was 10. A balanced logistic model with $C = 1$ predicted the sign of the final candidate margin from the same features. We report held-out R^2, mean absolute error and boundary AUROC, with 1,000 graph bootstrap replicates.

The order null retained the current-cutoff value and independently permuted all preceding values within each prompt. Fifty such null datasets were refitted and evaluated on the unchanged held-out graph groups. This destroys common history order while retaining each prompt's observed values and current state. The earliest cutoff whose real held-out R^2 exceeded the null 95th percentile was layer 24 in Qwen, layer 14 in Llama and layer 21 in Gemma. This analysis tests prospective information in a task-defined reduced trajectory. It is not a closed-form state equation, next-token correctness model or universal forecasting law.

Direct hidden-state chord geometry

The direct geometry audit used the 96-prompt controlled subset. States were normalized as $x_l = h_l/\|h_l\|_2$, step vectors were $s_l = x_{l+1} - x_l$, and the endpoint chord was $g = x_L - x_1$. Chord alignment was the mean cosine between s_l and g. Path length was $\sum_l [1 - \cos(x_l, x_{l+1})]$, endpoint distance was $1 - \cos(x_1, x_L)$, and detour was their ratio. Turning curvature was the mean angle between adjacent step vectors.

For Fig. 1, adjacent hidden-state change was $1 - \cos(x_l, x_{l+1})$. The dimensional participation fraction of a step was $(\sum_j s_{l,j}^2)^2/[d_m \sum_j s_{l,j}^4]$. A value near one indicates broadly distributed step energy, whereas a small value indicates concentration in fewer hidden dimensions. Means and standard errors were calculated within collapsed coordinate-audit condition family and checkpoint on the 96-prompt controlled subset.

For each checkpoint and each of 50 repeats, the null held the first and final states fixed and applied one random permutation to the intermediate layer order. The same permutation was applied to all 96 prompts within that checkpoint. This preserves the state set and endpoints while destroying observed layer order. The inferential unit was the checkpoint-level mean for one shared permutation; no prompt-wise p values were aggregated. Alignment gain was the real mean minus the shuffle mean. Detour and curvature reductions were

$$G_q = \log_2 \left(\frac{\mathrm{mean}(q_{\mathrm{shuffle}})}{\mathrm{mean}(q_{\mathrm{real}})} \right),$$

for $q \in \{\mathrm{detour}, \mathrm{curvature}\}$. For each metric, a null-standardized effect was the signed real-minus-null-mean difference divided by the standard deviation of the 50 checkpoint-level null means, oriented so that positive values favour observed order. A plus-one empirical one-sided p value was $(1 + n_{\text{null as or more extreme}})/(50 + 1)$; its minimum resolution was therefore $1/51$. The reported z values are standardized null separations, not Gaussian p values. These diagnostics are metric-specific and do not define a learned Riemannian metric or test an exact geodesic equation.

Independent task and random-initialization controls

The controlled lexical task contained 64 problems with candidate words drawn from seven categories. The controlled addition task contained 64 single-step problems. In each task, candidate order was counterbalanced and every problem supplied six matched prompt conditions. A fixed split assigned 44 problems to coordinate fitting and 20 to held-out evaluation. The task-specific coordinate was PC1 of training-problem clean-relative margin profiles and was applied without refitting; a one-coordinate logistic classifier was compared with 200 training-label shuffles. The numerical relation-graph ΔU axis was not reused.

For a standard non-binary benchmark, we used a deterministic 96-item ARC-Challenge validation audit subset whose answer labels were exactly A-D[26]. The subset comprised the earliest eligible rows in dataset order, a rule fixed without reference to model outputs to bound computation and preclude outcome-based selection. Original questions and all four choices were presented through each checkpoint's native chat template; no binary wrapper was added. This is a fixed audit set, not a new benchmark estimate. Lexical, addition and ARC geometry used the same normalized-state definitions and 50 endpoint-preserving shuffles as the relation-graph analysis; no model-specific window search was performed.

The random-initialization control used the exact Qwen2.5-1.5B, Llama-3.2-1B and Gemma-2-2B configurations and native tokenizers. Three models per architecture were instantiated with seeds 1701, 1702 and 1703. Embeddings, transformer blocks, normalization parameters and output heads were newly initialized; no trained tensor was loaded. Each random state and its matched trained checkpoint used the same 96 raw relation-graph prompts and 50 endpoint-preserving shuffles with null seed 20260717. The separately seeded Fig. 4 null gives slightly different trained gain estimates although the real trajectories are identical. This within-architecture comparison separates architecture-compatible order effects from training-specific organization for the tested configurations and prompt subset; it does not identify a universal Transformer law.

Frozen closure audits

The closure protocol fixed the predecision cutoffs, five-fold graph partition, metrics, nulls and multiplicity families before execution. For the coordinate competition, R_l was the clean-minus-conflict output-logit margin at predecision layer l, and dR_l was its clean-condition-matched version in the same graph group. Each training fold fitted scaling, PCA and a class-balanced multinomial logistic classifier. The compared features were the current or recent R_l and dR_l, a one-dimensional PC1 score ΔU_{pre} from the complete training-fold dR history, the complete standardized dR history, and a 32-component training-fold PCA of the cutoff hidden state. No post-cutoff state, final token, correctness label or held-out statistic entered the construction. The primary contrasts used 100,000 graph-level sign flips and Benjamini-Hochberg adjustment across nine checkpoint-by-contrast tests (Supplementary Methods Table 9).

For the direction null, the released held-out gate, layer schedule, action weights, local operator and precursor components were retained exactly. A random direction was sampled at each intervention layer from the empirical training-update subspace, projected orthogonally to the structured component and normalized to the same component norm. Fifty seeds per checkpoint preserved the complete gate pattern, action schedule, layer placement and full-vocabulary endpoint. The primary specificity was closure strict conflict-to-clean crossings minus non-closure full-vocabulary top-1 changes. The three checkpoint tests form one multiplicity family (Supplementary Methods Table 10).

For new-graph geometry, 24 disjoint graph groups and ten fixed conditions per group supplied the same 240 raw prompt strings to Qwen2.5-1.5B and Gemma-3-12B-it-QAT-Q4_0. All native block outputs were captured. Fifty shared permutations retained the first and last block states and permuted only the intermediate order for every prompt in a repeat. Chord alignment was the primary outcome; detour and turning curvature were supporting metrics. The two checkpoint-level chord tests form a separate multiplicity family (Supplementary Methods Table 11). The larger Gemma checkpoint was loaded through the pinned native capture runtime; its quantization and model digest are in Extended Data Table 1 and Source Data.

Falsification-first scaffold, order and metric controls

The cross-task random-map test instantiated two additional seeds (1801 and 1802) for each architecture and reused each fixed map across 96 lexical, 96 addition and 96 ARC prompts. Cross-half and first-to-last-quarter unbiased linear CKA were compared with 199 condition- and answer-stratified lag-1 donor, contiguous four-step donor and prompt-fingerprint-preserving donor nulls. A condition passed only when both CKA measures exceeded the 95th percentile of all three nulls at empirical $P \leq 0.05$. This tests relation preservation by a fixed random architecture, not semantics, native-order privilege or function.

The order-function dissociation used a trained Qwen residual-execution cache for 48 prompts under native, reverse and one frozen permuted block order. An eight-prompt reconstruction audit reproduced native full-vocabulary top-1 exactly and candidate margins within 0.05 before the cache was used. Geometry was the paired change in unit-state future-chord alignment. Functional endpoints were full-vocabulary top-1 agreement, candidate-choice agreement, Jensen-Shannon divergence and native-token rank; 10,000 paired bootstraps and 100,000 group sign flips quantified uncertainty. Passing means only that greater alignment is insufficient for native functional fidelity.

The incremental path-memory audit compared ordered candidate-margin prefixes with the current margin, a fold-fitted 32-component current hidden state, unordered prefix summaries and 199 within-prompt order shuffles. Five-fold problem-group splits covered lexical and addition tasks at 25%, 50% and 75% depth in each checkpoint; a separate mixed-arithmetic task repeated the same nine model-by-horizon tests. Benjamini-Hochberg correction was applied within contrast family. A condition passed only if ordered history exceeded all four controls at $q \leq 0.05$. This audit is separate from the relation-graph forecast, which retains a positive but task-bounded result.

The Llama radial audit used consistent block-output residual states from four original and three independent tasks. For $h_l = r_l u_l$, direction-only paths set r_l constant and norm-only paths held u_l fixed. A frozen dose series used $h_l(\gamma) = (r_l/r_0)^\gamma u_l$ for $\gamma \in \{0, 0.25, 0.5, 0.75, 1, 1.25\}$, with 199 layer-conditioned recipient-norm donor walks at each value. The statistic was strongly radial-sensitive, but the deliberately strict binary sufficiency gate failed; no functional or semantic conclusion follows.

Training-time component and coordinate audits

We used one documented EleutherAI Pythia-160m training run at step 0 and step 143,000. On a frozen 53-prompt single-token lexical cloze set, all 16 combinations of step-0 and trained input embeddings E, transformer blocks B, final normalization N and untied output head W were evaluated. Primary endpoints were clean-versus-conflict pair accuracy and mean candidate margin. Prompt-paired uncertainty used 5,000 bootstraps; factorial contrasts used the complete orthogonal 2^4 design. Component boundaries and prompts were fixed before outcomes were inspected. Hybrid states are perturbational controls, not natural training checkpoints.

The learned-coordinate audit used the same 53 prompts at training steps 0, 128, 1,000, 4,000, 16,000, 64,000 and 143,000. The model-native candidate coordinate was $c_{p,l} = \cos(h_{p,l}, W_{c_p}) - \cos(h_{p,l}, W_{e_p})$; mean prompt-wise Spearman correlation with depth was compared with 999 endpoint-preserving interior-layer shuffles. For the category coordinate, clean answer words rather than prompt rows were divided into three deterministic folds. A 16-component training-fold PCA and fixed-regularization multinomial logistic regression were applied to every layer of held-out words. We compared a coordinate refitted within each checkpoint with the final-checkpoint coordinate transferred without refitting to earlier checkpoints, and used 199 category-label permutations at step 0 and the final checkpoint. The explicit category word made within-checkpoint random features linearly decodable at initialization; that negative control is retained rather than interpreted as learned semantics.

Readback-centre geometry and residual diagnostics

The auxiliary readback-centre analysis used normalized vocabulary-neighbourhood centre trajectories rather than raw hidden states. Its endpoint chord, cosine-distance path, detour and turning metrics used an endpoint-preserving permutation of intermediate readback-centre layers. This analysis is therefore reported in Extended Data as a cross-observable diagnostic and is not merged with the direct hidden-state estimand.

Residual diagnostics compared each observed update with its chord-directed reference and summarized components parallel and perpendicular to the reference direction. These are reduced-order residual proxies. They test whether the remainder retains classification or ΔU-prediction information, but are not an exhaustive tangent-normal decomposition.

Mechanism-conditioned intervention

Three additive-state controls used prototype- or linear-coefficient-derived directions; a subsequent additive-flow control used a flow direction. Because these controls did not provide robust mechanism-specific selectivity, the final intervention changed the actuator to a mechanism-conditioned local transition operator. At intervention layer l, a common principal-component basis was fitted to the concatenated current and next hidden states from training graphs only, with dimension $r_l = \min(128, 2n_{\text{train}} - 1, d_m)$.

For stable and closure rows separately, ridge regression estimated $(A_l^{(k)}, b_l^{(k)})$ by

$$\underset{A,b}{\operatorname{argmin}} \sum_{i\in\mathcal{T}_k} \|z_{i,l+1} - Az_{i,l} - b\|_2^2 + 10\|A\|_F^2,$$

where z denotes the shared PCA coordinate and $k \in \{\text{stable}, \text{closure}\}$. No held-out graph entered either fit. The stable-row prediction was inverse-transformed to hidden-state space to define $\hat{F}_l^{\text{stable}}$. For realized block output $h_{p,l+1}$, its proposal was applied as

$$\tilde{h}_{p,l+1} = (1-\alpha_{p,l})h_{p,l+1} + \alpha_{p,l}\hat{F}_l^{\text{stable}}(h_{p,l}),$$

followed by the order-precursor update

$$h_{p,l+1}^{\text{int}} = \tilde{h}_{p,l+1} + \beta_{p,l}\,\text{sd}(\tilde{h}_{p,l+1})d_l.$$

Here d_l is the unit ridge-coefficient direction for ΔU, fitted on training graphs; α controls operator interpolation and β controls the scaled precursor update. The operator step was applied before the precursor step. The action grid was $\{(0,0),(0.1,0.1),(0.2,0.2),(0.3,0.3),(0.3,0),(0,0.3),(0.3,0.2),(0.2,0.3)\}$. The fixed-combination baseline used $(0.3, 0.3)$.

The intervention-class classifier, called the mechanism classifier in the analysis code and figure labels, used three prompt-condition labels fixed in the ten-condition control panel. Stable comprised clean, redundant, irrelevant and paraphrase conditions; competition comprised weak-distractor, balanced-competition and direct-conflict conditions; closure comprised closure-update, closure-override and exception-override conditions. These are controlled intervention families, not correctness labels or trajectory-derived dynamical phases. Its nine input features were predicted candidate-path separation and its magnitude, a companion dominance estimate, velocity magnitude and standardized velocity, normalized layer depth, a precursor-layer indicator, and baseline separation and dominance values. A 250-tree random forest used maximum depth 7, minimum leaf size 6 and class-balanced bootstrap weights. The policy classifier added three intervention-class probabilities and three predicted-class indicators, giving 15 inputs, and used a 200-tree random forest with maximum depth 6, minimum leaf size 8 and the same weighting rule. Both used standard weighted-Gini tree splits. There were no neural hidden layers, gradient optimizer or policy-gradient objective.

Policy supervision was obtained by evaluating all eight (α, β) actions on training graphs. For closure prompt p, the target action maximized $\Delta U_p^{(a)} - \Delta U_{p,\text{base}}$; for stable and competition prompts, it minimized the absolute shift. The policy forest then classified the 15-dimensional held-out layer features into the resulting discrete action labels. Policy labels encode (α, β); none denotes $(0, 0)$.

Classifiers used a 70:30 group-aware split by relation graph on layer-expanded rows. Qwen and Gemma used 3,350 training rows from 670 policy-training prompts and Llama used 2,680 rows because of its shorter layer stack; 290 prompts formed the held-out test partition. Rows from one graph could not occur in both partitions. In a mechanism-label null, training mechanism labels were permuted before refitting the mechanism classifier and downstream policy. In a policy-label null, the action labels were permuted before refitting the policy while retaining the real mechanism classifier. The joint null independently permuted both label sets and refitted both classifiers; test features, graph split, transition operators and evaluation code were unchanged. Each null family was repeated 50 times. Final-answer correctness was not a classifier target.

For checkpoint m and condition c, specificity was

$$s(m,c) = \mathrm{mean}_{i\in S_{\text{closure}}} |\Delta U_i^{(c)} - \Delta U_{i,\text{base}}| - \mathrm{mean}_{i\in S_{\text{nonclosure}}} |\Delta U_i^{(c)} - \Delta U_{i,\text{base}}|.$$

Fig. 5 reports real and matched fixed-combination specificity; their difference is the real-versus-fixed gain. Shuffle-null separation is reported against the joint shuffle distribution. For visualization only, the pseudo-log transform is $f(x) = \operatorname{asinh}[x/(2\sigma)]/\ln 10$, with $\sigma = 0.08$; it is approximately linear near zero and logarithmic at larger magnitudes. It does not change the underlying specificity calculation.

The candidate-choice boundary analysis paired the no-intervention and mechanism-aware-policy rows for each of 290 held-out prompts per checkpoint. The clean-candidate margin was the final clean-candidate logit minus the final conflict-candidate logit. Changes in margin, ΔU and the binary choice between these two candidates were calculated within prompt. Ninety-five per cent intervals for the paired descriptive contrasts resampled the 29 held-out relation graphs with replacement for 5,000 replicates using seed 20260714. Exact paired-flip tests compared conflict-to-clean with clean-to-conflict flips.

For the displacement-to-boundary bridge, a five-fold graph split shared across checkpoints cross-fitted model- and regime-specific affine maps $\delta M = a_{m,r} + b_{m,r}\delta\Delta U$, where δM is the policy-induced candidate-margin change. The predicted post-policy margin was $M_0 + \widehat{\delta M}$. On prompts with $M_0 < 0$, crossing meant that the observed policy margin was non-negative. Nested graph-held-out logistic calibration converted predicted post-margin to crossing probability. Continuous prediction and crossing metrics used 1,000 graph-bootstrap replicates. Final correctness labels did not enter either model. Clean denotes the unperturbed-chain candidate and is not necessarily the rule-consistent answer under closure; this analysis measures crossing between the two declared candidates, not unrestricted generated-answer correctness.

Confidence-gated boundary control

The output-boundary extension was run independently in Qwen, Llama and Gemma on the same seed-42 graph split: 67 training graphs (670 prompts) and 29 held-out graphs (290 prompts, including 87 closure prompts). Intervention layers were 15-19, 8-11 and 13-17, respectively. In every checkpoint, the principal-component direction defining ΔU, local transition operators, precursor directions, mechanism classifier, action labels and boundary policy were fitted using training graphs only. Candidate token identities entered the paired-margin objective; the model's generated answer did not enter fitting or gate construction.

The expanded action library was

$$\begin{aligned}\mathcal{A} = \{&(0,0),(0.3,0.3),(0.6,0.3),(0.6,0.6),(0.9,0.3),\\ &(0.9,0.6),(0.9,0.9),(1.0,0.6),(1.0,0.9),(1.0,1.2)\}.\end{aligned}$$

For a training closure prompt, action utility was the clean-minus-conflict margin shift plus a 5-unit crossing bonus minus $0.03(\alpha^2+\beta^2)$; for non-closure prompts it was the negative absolute margin shift minus $0.10(\alpha^2+\beta^2)$. Thus $\alpha \le 1$ interpolated to, but did not extrapolate beyond, the fitted transition proposal. The policy used the same 15 pre-intervention features as the internal-control analysis.

For prompt p, let $\pi_{p,l}(k)$ be the mechanism posterior at intervention layer l. We defined

$$\bar{\pi}_p = |L_I|^{-1} \sum_{l \in L_I} \pi_{p,l}(\text{closure}), \qquad \bar{H}_p = |L_I|^{-1} \sum_{l \in L_I} \frac{-\sum_k \pi_{p,l}(k) \log \pi_{p,l}(k)}{\log 3},$$

and opened the gate only when $\bar{\pi}_p \geq 0.50$ and $\bar{H}_p \leq 0.90$. These thresholds were fixed in the experiment protocol before held-out execution; no threshold sweep or held-out outcome tuning was performed. They should therefore be read as one declared operating point, not an estimated optimum. Gate-closed actions were set to $(0, 0)$. The primary behavioural endpoint was a strict change from the conflict token to the clean token as the full-vocabulary top-1 continuation. Non-closure top-1 change and intervention-layer norm ratios were damage guards.

For the safety-matched null, the complete model-specific action vector was permuted without fixed points among gate-open prompts within each closure condition. Each of 50 permutations preserved the gate-open set, zero actions outside the gate, and every condition-by-layer (α, β) multiset. Strict conflict-to-clean top-1 changes were counted relative to each prompt's unperturbed baseline. The one-sided empirical value was $(1 + n_{\text{null} \geq \text{real}})/(50 + 1)$. The real counts did not exceed the matched-null 95th percentile in any checkpoint. This null tests action-to-trajectory assignment, not whether the gated actuator family can cross the candidate boundary.

For component attribution, the gate-open set, thresholds, graph split, operators, precursor directions and all training fits were frozen before execution. The operator-only control retained each gated policy value of α and set $\beta = 0$; the precursor-only control retained β and set $\alpha = 0$. A uniform control applied the pre-existing maximum library action $(1.0, 1.2)$ to every gate-open prompt and zero elsewhere. Negative and positive references were rerun and reproduced all 290 previous top-1 token IDs exactly in every checkpoint. Operator-only strict crossings were 17/87, 14/87 and 8/87; precursor-only crossings were 0/87, 1/87 and 0/87; and uniform-maximum crossings were 22/87, 15/87 and 10/87. Non-closure top-1 changes were 0/203, 2/203 and 0/203 for operator only, and 0/203, 2/203 and 1/203 for the uniform control. All intervened states were finite and the maximum recorded state-norm ratio was below 1.10. These ablations attribute most leverage to the gated local transition operator within the tested actuator; they do not establish a universal decomposition.

Frozen mixed-arithmetic actuator-transfer test

The transfer test contained 64 unique addition, subtraction and multiplication expressions with answer labels from zero to twenty. Each problem supplied clean, paraphrase, untrusted-conflict, asserted-conflict, explicit-correction and final-recheck prompts. The first two were non-target stable prompts, the middle two non-target competition prompts and the final two target correction prompts. All six rows from a problem were assigned together by a frozen split (44 training and 20 held-out problems; seed 20260714). The arithmetic-consistent candidate is correct only within this exact one-word task contract.

The relation-graph intervention windows, ten-action library, operator regularization, classifier settings, gate thresholds and 5% collateral ceiling were frozen. Task-specific coordinates, numerical transition operators, precursor directions and action policies were fitted on training problems only; this tests protocol transfer after training-only refitting, not zero-shot transfer of numerical parameters. The primary interface ended a raw prompt immediately before its one-word continuation, matching the original actuator assay. All candidate labels passed an exact-prefix and single-token continuation audit in all three tokenizers. A native-chat pilot was excluded from the primary endpoint because Llama began all 120 held-out responses with free text rather than either candidate, demonstrating that a candidate-token boundary must be valid for the evaluated interface.

Strict eligibility required the unperturbed full-vocabulary top-1 token to equal the distractor candidate on a held-out correction prompt. The primary guarded actuator was compared with no intervention, its ungated form, a training-selected fixed action, operator-only, precursor-only, uniform-maximum and 50 address-reassignment nulls per checkpoint. Each address null reassigned complete action bundles across held-out prompts while preserving the model-specific action-dose multiset and number of open gates. Specificity was the strict target-crossing rate minus the non-target top-1-change rate. Confirmation required at least two checkpoints with at least five eligible prompts, positive median candidate-margin movement and at least one strict crossing; at least two checkpoints below the collateral ceiling; and pooled specificity above the address-null 95th percentile. Qwen yielded 1/14 crossings, median margin shift 0.539 and 1/80 non-target changes; Llama yielded 1/27, -0.070 and 0/80; Gemma yielded 0/4, 0.094 and 0/80. Pooled specificity was 0.0403 versus a null 95th percentile of -0.0111 (one-sided empirical $P = 1/51$). Independent recomputation reproduced every primary summary and all 150 null action-dose multisets. The two-checkpoint positive-direction gate failed, fixing the verdict as partial transfer rather than cross-task output control.

Statistics and reproducibility

The analysis units, group splits, prompt subsets, windows and nulls are summarized in Supplementary Methods Tables 8-11. The claim-family register separates heterogeneous tests by their statistical unit and null; no single study-wide adjusted P value is claimed. Main comparisons are descriptive estimates and matched-null z scores; no single background citation is used to support a core mechanistic result. Processed panel tables are mapped by the global Source Data manifest. Figures were rendered deterministically in R using `ggplot2`, `patchwork` and related packages and exported as vector PDF/SVG and 600-dpi TIFF. Scientific panels were generated from retained data and code, not by a generative image system.

The direct hidden-state geometry analysis used seed 20260610. A fixed-seed rerun reproduced the hashes of the controlled-subset manifest, prompt-level metrics and summary tables. Exact checkpoint revisions and extraction rules are recorded in Extended Data Table 1; classifier features, subsets and null construction are recorded in Supplementary Methods Table 8, and the claim-boundary audit is Supplementary Methods Table 4. Repository accessions and source mappings are reported below.

Research independence and resource provenance

This study was conducted independently and was not an internal project of any listed organization. No employer data, computing infrastructure, internal systems, funding, research facilities or personnel support were used, and the work did not involve organizational review or approval. All local computing, storage, software subscriptions and other research expenses were provided or paid for personally by H.Z.; public model checkpoints and software were accessed under their stated licences. Y.W., R.S. and W.C. participated in their personal capacities as independent research collaborators. Affiliations identify the authors and do not imply employer sponsorship, endorsement or responsibility for the work.

AI-assisted research workflow and human oversight

Generative AI tools, including GPT (OpenAI), Gemini (Google), Doubao (ByteDance) and Kimi (Moonshot AI), were used as interactive software for archival retrieval, code scaffolding and debugging, batch-analysis planning, data organization, deterministic R plotting scripts, consistency audits and language editing. H.Z. formulated the scientific questions, selected and revised the empirical process object, specified controls and acceptance criteria, interpreted positive and negative results, set claim boundaries and approved all reported analyses, figures and text. Y.W., R.S. and W.C. provided the domain and methodological support described in the Author contributions statement, and all authors reviewed the manuscript. Analyses were executed locally against the stated checkpoints and source data; generated proposals and code were checked against retained outputs, null controls and source mappings. AI outputs were treated as unverified candidate hypotheses or operational drafts, not as evidence. AI systems were not authors and made no autonomous scientific or editorial decisions. No scientific conclusion was accepted solely from AI output, and no publication figure was produced by an image-generation system. Timestamped records document the compressed N=1 workflow, but without a no-AI counterfactual they support workflow provenance rather than a causal estimate of productivity. The task-level workflow and responsibility matrix are reported in Supplementary Methods.

Data availability

Processed panel- and row-level tables are archived under the Zenodo Source Data concept DOI `https://doi.org/10.5281/zenodo.21317355` and mirrored in the paper-specific public branch at `https://github.com/zhaoheng1990-debug/Cognitive_Theory_And_LLM_Phasemap/tree/high-dimensional-layer-state-trajectories`. The public repository release v0.55 adds the frozen mixed-arithmetic transfer audit and retains the target-excluded transition-prediction audit, prospective functional-window boundary test, analysis-label crosswalk, actuator-component attribution and submission-ready Source Data manifest. Prompt and checkpoint metadata, random seeds, split records, null specifications, figure contracts and file-level checksums are included. Raw hidden-state arrays are large derivatives of third-party checkpoints and are not redistributed; prompts, checkpoint revisions and extraction scripts required to regenerate them are provided subject to checkpoint access, storage and licensing constraints. Model weights are not redistributed.

Code availability

Analysis scripts, deterministic R figure scripts, environment records, smoke tests and portable path templates are archived under the Zenodo Code concept DOI `https://doi.org/10.5281/zenodo.21317440` and mirrored in the paper-specific public branch at `https://github.com/zhaoheng1990-debug/Cognitive_Theory_And_LLM_Phasemap/tree/high-dimensional-layer-state-trajectories`. The public repository release v0.55 adds a frozen mixed-arithmetic transfer runner and independent verifier alongside the earlier neutral-named entry points and methodology-closure audits. Workstation-specific paths and internal development identifiers are removed or replaced with repository-relative arguments while provenance is retained in the private audit record. Full model reruns require the listed checkpoints and any upstream datasets that cannot be redistributed.

Acknowledgements

I (H.Z.) am deeply grateful to my family. The initial inspiration for this research arose while I was thinking about how to educate my son. I thank my parents, my wife and my son for supporting my decision to step away from my primary professional work and devote sustained time to an independent research interest. My wife also covered the subscription costs for GPT and Gemini used during the project.

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Author contributions

H.Z. conceived the research, defined the questions, formulated the hypotheses developed during the investigation, designed the experiments, established the evidential tests and revised the interpretations in response to the results. Y.W., an artificial-intelligence specialist, contributed domain knowledge and methodological guidance throughout the research. R.S., a researcher in operations research and mathematics, contributed mathematical-method support to parts of the study. W.C. participated in discussions of the research questions and provided advice and conceptual inspiration from medical, neuroscience and cognitive-science perspectives. Y.W., R.S. and W.C. contributed in their personal capacities as independent research collaborators and did not provide employer data or resources. All authors reviewed the manuscript and approved the submitted version. AI-assisted operations are disclosed in Methods and do not constitute authorship.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information is available for this paper. Correspondence and requests for materials should be addressed to Heng Zhao (zhaoheng1990@gmail.com).

Extended Data tables

Extended Data Table 1 | Model, checkpoint and hidden-state extraction metadata.

Family	Exact checkpoint and revision	Layers / hidden size	Extraction
Qwen	Qwen/Qwen2.5-1.5B-Instruct, 989aa7980e4cf806f80c7fef2b1adb7bc71aa306	28 / 1,536	Last non-padding token from block output $l + 1$; embedding excluded
Llama	meta-llama/Llama-3.2-1B-Instruct, 9213176726f574b556790deb65791e0c5aa438b6	16 / 2,048	Last non-padding token from block output $l + 1$; embedding excluded
Gemma	google/gemma-2-2b-it, 299a8560bedf22ed1c72a8a11e7dce4a7f9f51f8	26 / 2,304	Last non-padding token from block output $l + 1$; embedding excluded
Gemma new-graph confirmation	Gemma-3-12B-it QAT Q4_0 GGUF; full model digest in Source Data run metadata	48 / 3,840	Last-token state captured from every block output through the pinned native capture source; QAT Q4_0 quantization

Figures and legends

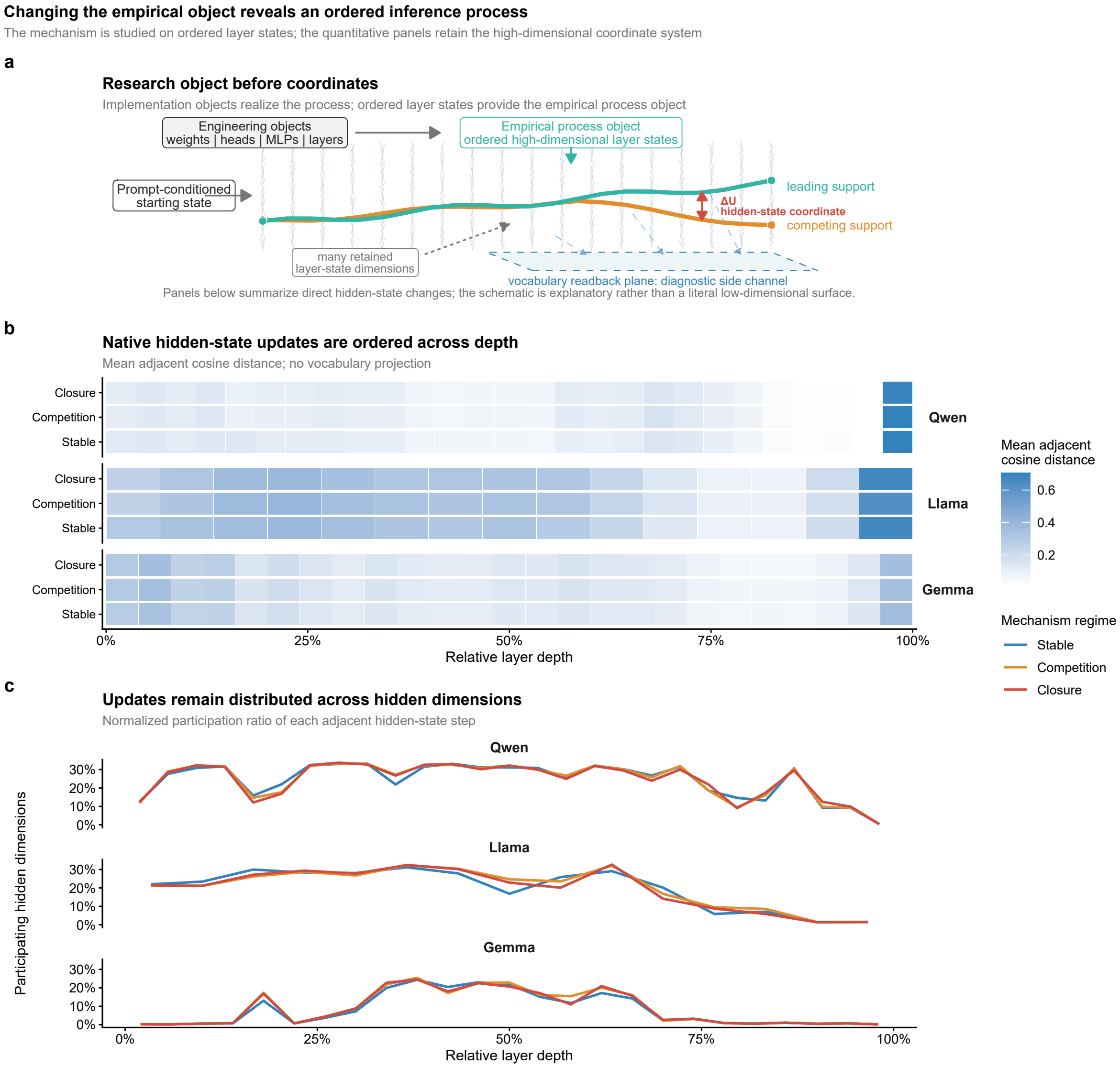


Fig. 1 | Changing the empirical object reveals an ordered inference process. a, Distinction between implementation objects, including weights, attention heads, MLPs and layers, and the empirical process object used here: the ordered high-dimensional layer-state trajectory. b, Mean adjacent cosine distance between normalized hidden states across relative depth and collapsed prompt-condition families. c, Normalized participation ratio of adjacent hidden-state steps, reporting the fraction of hidden dimensions over which step energy is distributed. Panels b and c use direct hidden states without vocabulary projection. The condition families are predefined analysis labels, not trajectory-derived phases. Layer depth is an ordered evolution coordinate, not physical time. The schematic does not depict a literal low-dimensional surface. See Source Data Fig. 1 and Source Data manifest; model and extraction metadata are in Extended Data Table 1.

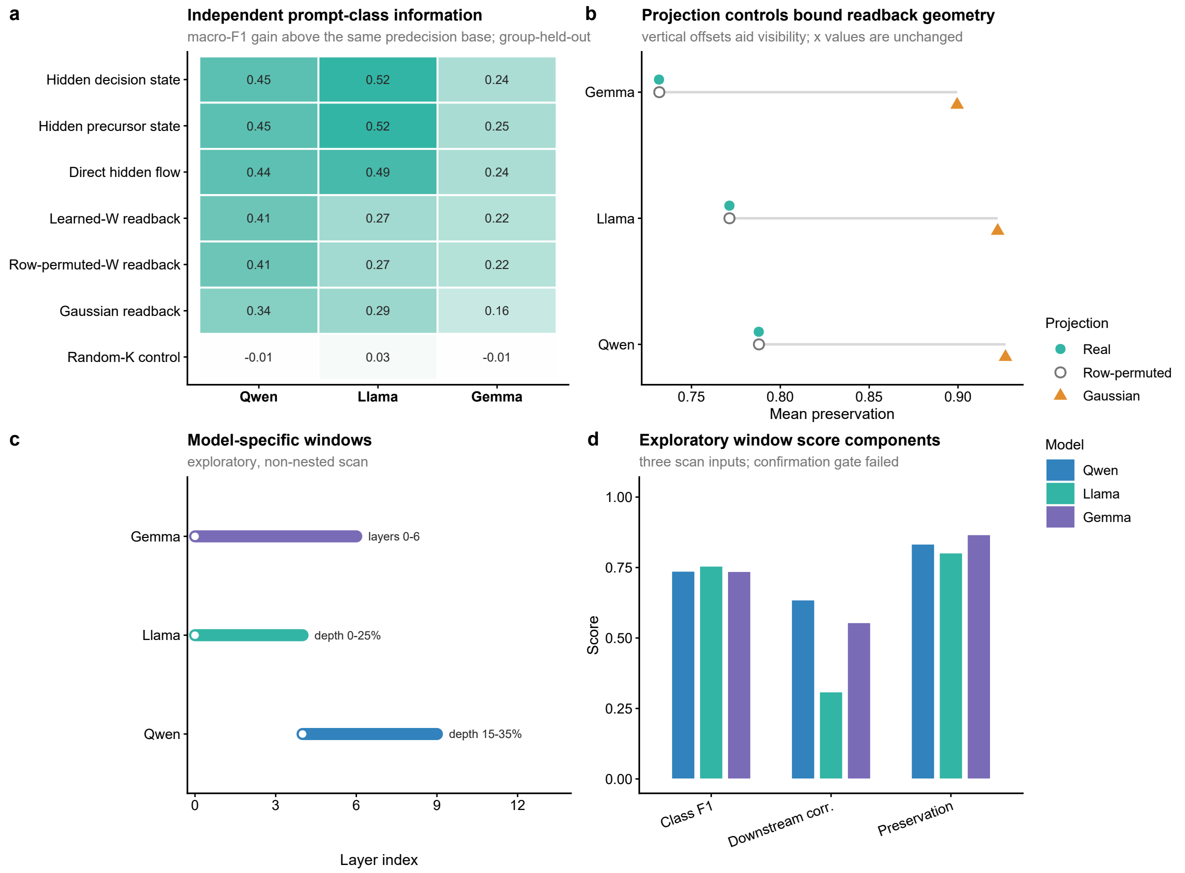


Fig. 2 | Coordinate audit and exploratory functional registration. a, Group-held-out prompt-condition-class macro-F1 increment above the same predecision base; positive values indicate additional information. Reduction was fitted within each training fold. b, Mean readback-geometry preservation under real, row-permuted and Gaussian projections; vertical offsets only separate overlapping points. c, Discovery-scan windows as layer or relative-depth intervals. d, The three exploratory scan components: prompt-class F1, downstream correlation and clean-relative preservation. The construction-aligned profile is excluded from a because it shares its matrix with ΔU (Extended Data Fig. 2). Readbacks remain diagnostics. A prospective 24-graph test failed the joint three-checkpoint recurrence gate, so the windows remain exploratory (Supplementary Methods Table 2). See Source Data Fig. 2 and Source Data manifest.

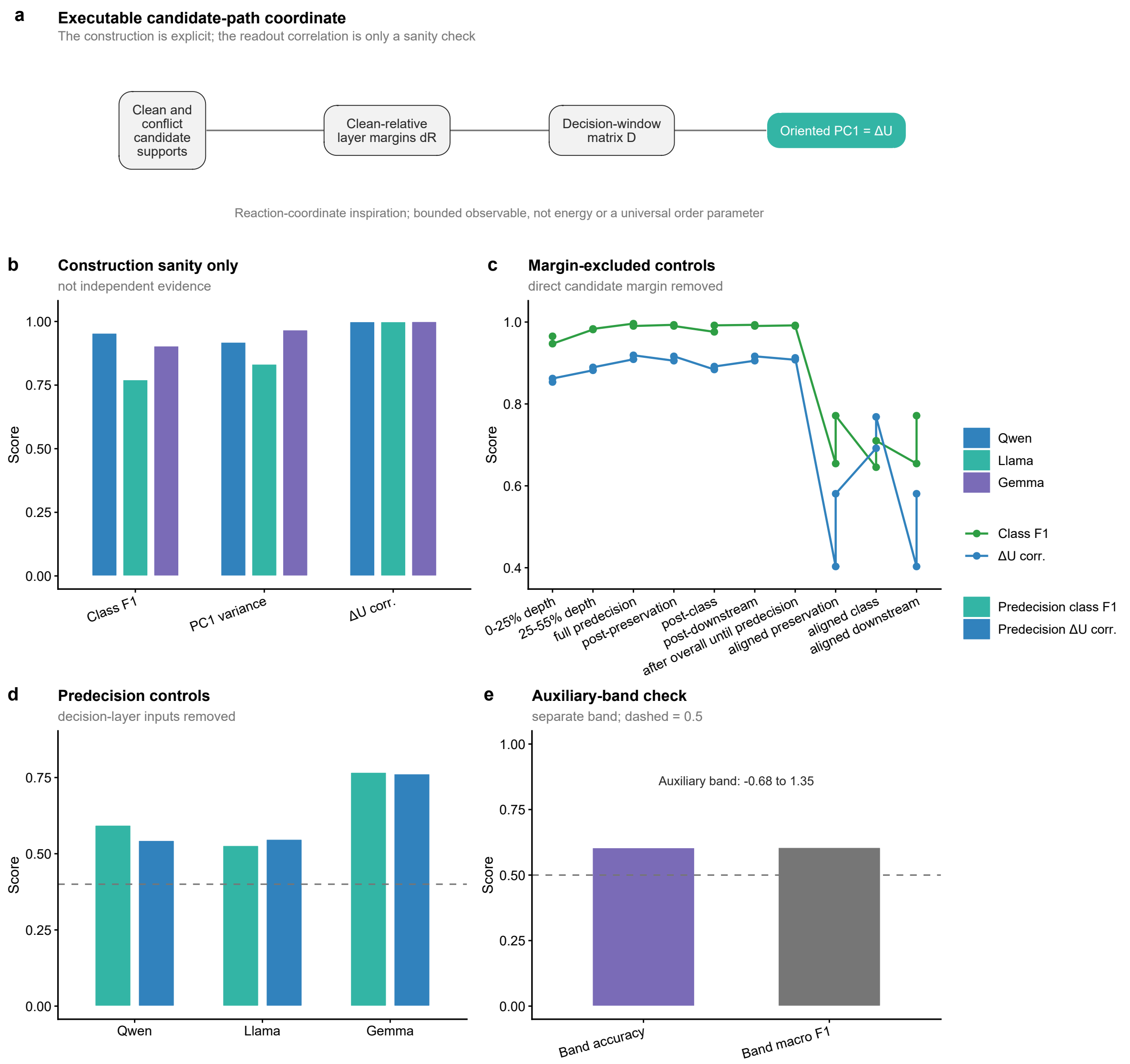


Fig. 3 | Construction and bounds of candidate-path separation. a, Candidate supports define clean-relative layer margins dR, the decision-window matrix D and oriented PC1 score ΔU. b, Prompt-class F1, PC1 variance and the construction-aligned ΔU-dominance check; this construction sanity check is not independent evidence. c, Direct candidate-margin inputs are excluded. d, Decision-layer inputs
875 are excluded. In c and d, green is class F1, blue is ΔU association, and higher values indicate greater retention of the original diagnostic. e, Auxiliary-band accuracy and macro F1. Dashed lines mark 0.4 in d and 0.5 in e. ΔU is a bounded candidate-path separation observable, not an energy or universal order parameter. See Source Data Fig. 3 and Source Data manifest.

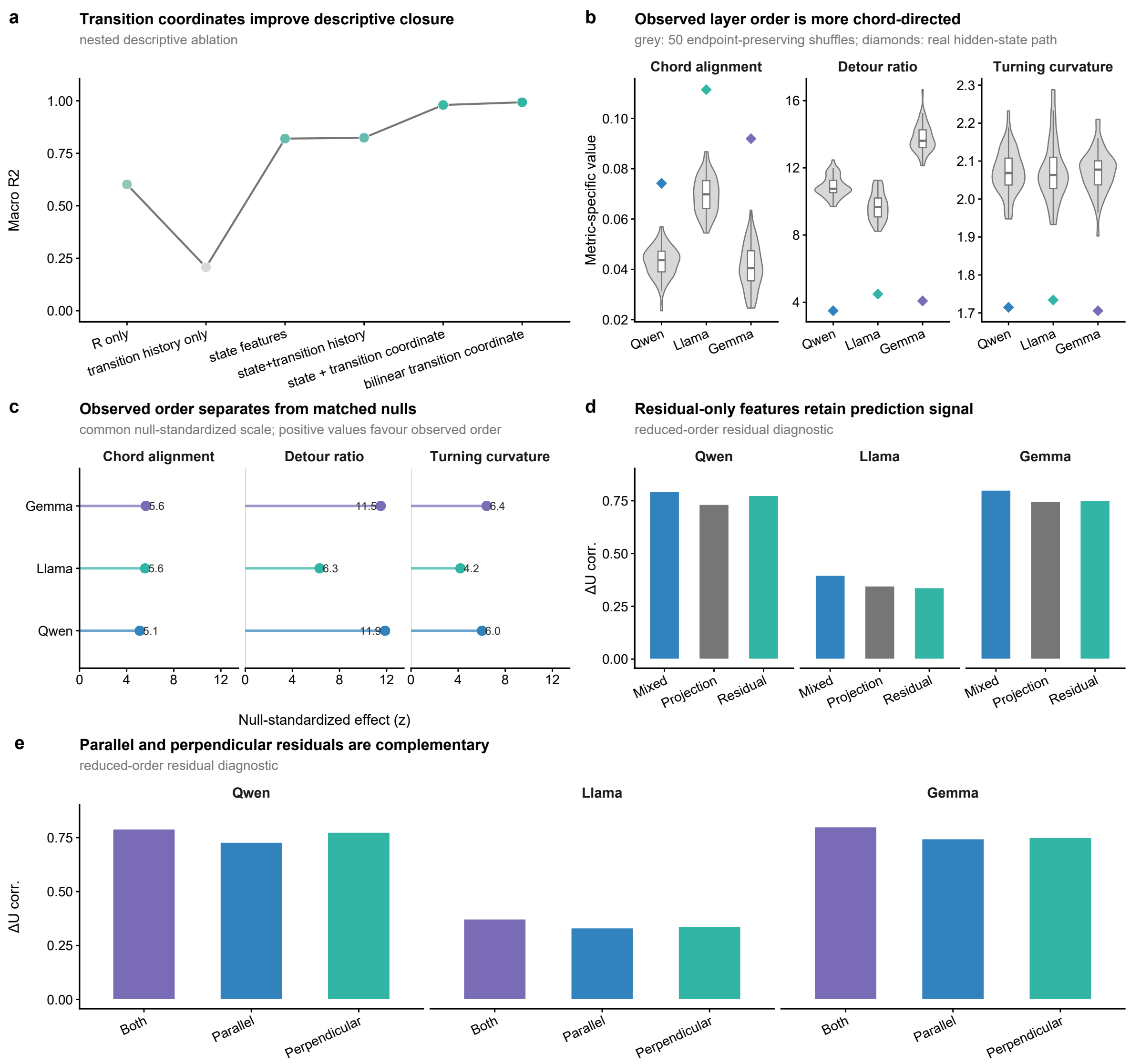


Fig. 4 | Realized transitions and layer order reveal structured flow. a, Descriptive closure for state-only, previous-transition history, state plus O_l^{cont}, and bilinear realized-transition models. When target-transition information enters O_l^{cont}, the result is descriptive rather than predictive. b, Hidden-state chord alignment for real trajectories and 50 endpoint-preserving layer-order shuffles. c, Null-standardized effects for chord alignment, detour and turning curvature on a common scale; positive values favour observed order. d, Residual-only prediction. e, Parallel and perpendicular residual proxies; these proxies are not an exhaustive tangent-normal decomposition. The geometry is metric-dependent and is not an exact-geodesic test. The transition-prediction boundary is shown in Extended Data Fig. 4. See Source Data Fig. 4 and Source Data manifest.

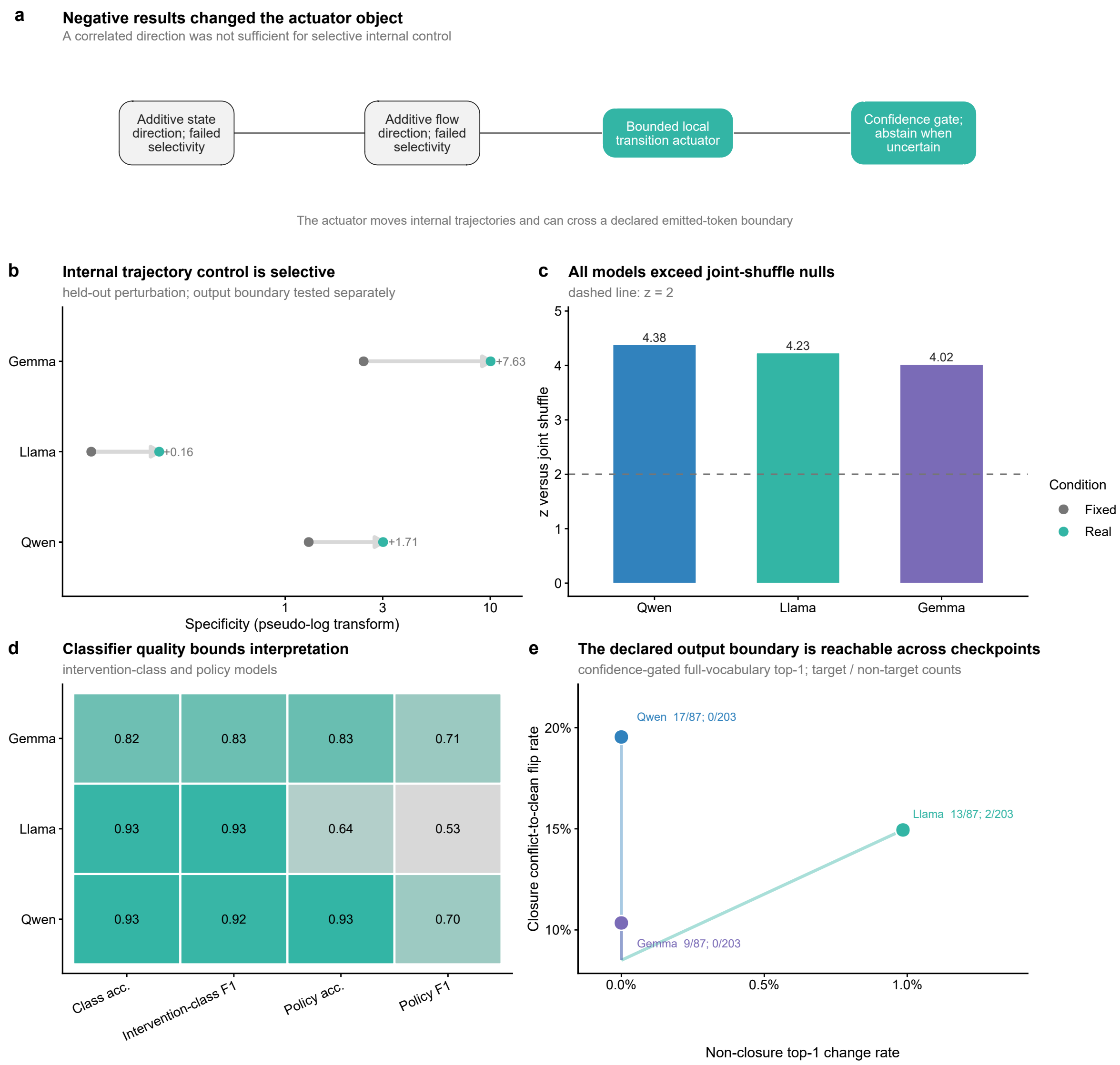
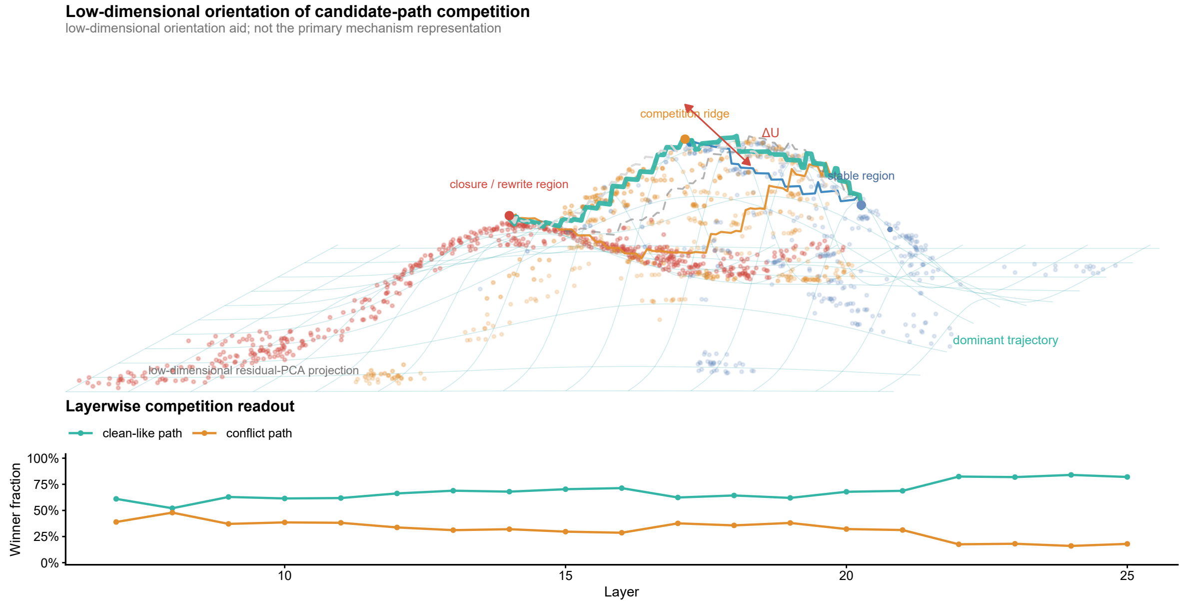
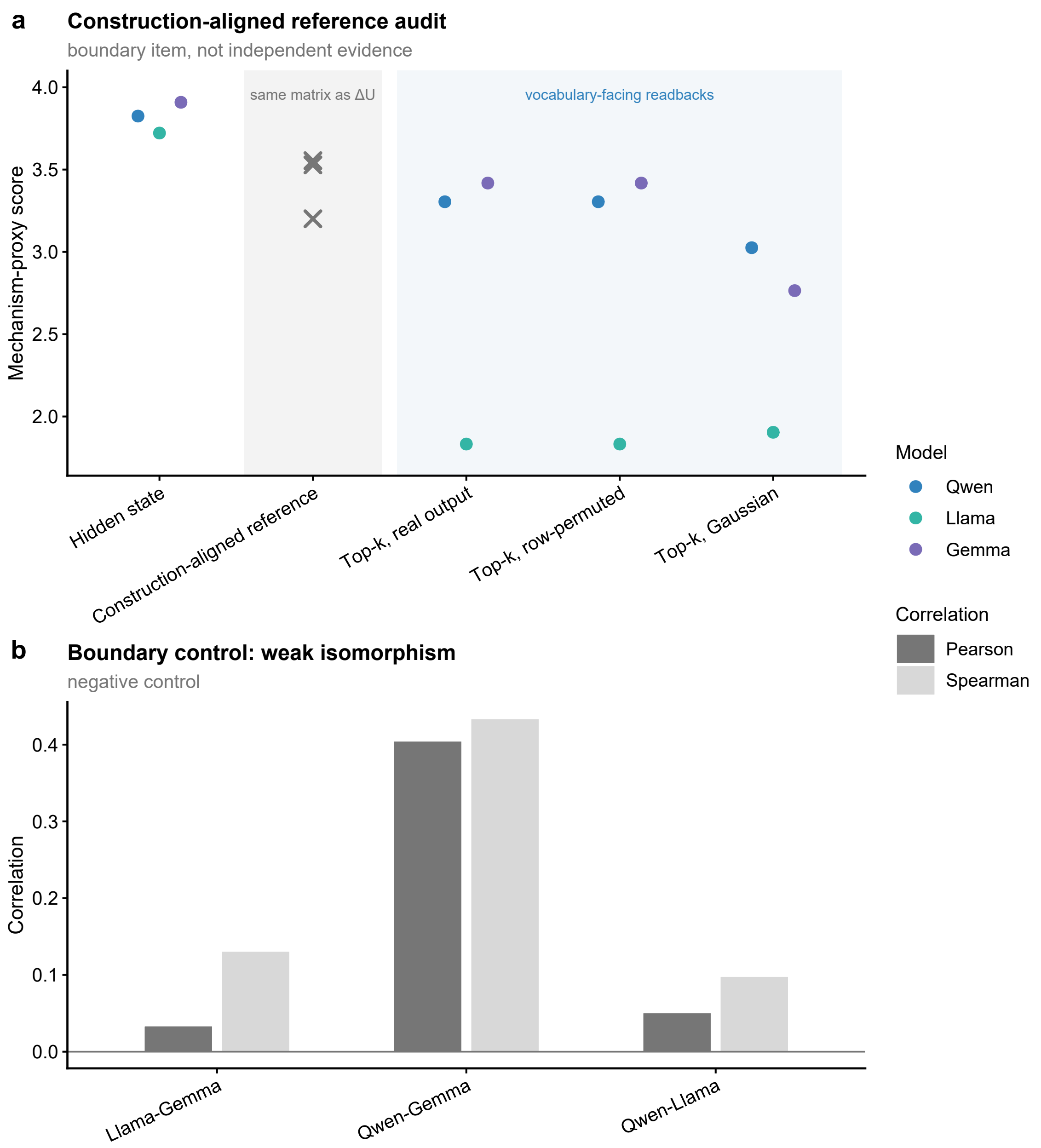


Fig. 5 | A bounded local actuator controls trajectories and a declared output boundary. a, Additive failures, bounded actuator and gate. b, Real-versus-fixed internal specificity. c, Real policy versus 50
890 joint-label shuffles. d, Classifier quality. e, Strict full-vocabulary conflict-to-clean crossings among 87 held-out closure prompts and collateral changes among 203 non-closure prompts: Qwen, 17/87 and 0/203; Llama, 13/87 and 2/203; Gemma, 9/87 and 0/203. Clean is provenance-defined; counts do not measure accuracy. See Extended Data Fig. 6f for action reassignment, Supplementary Methods Table 5 for non-crossings, and Source Data Fig. 5 and manifest.

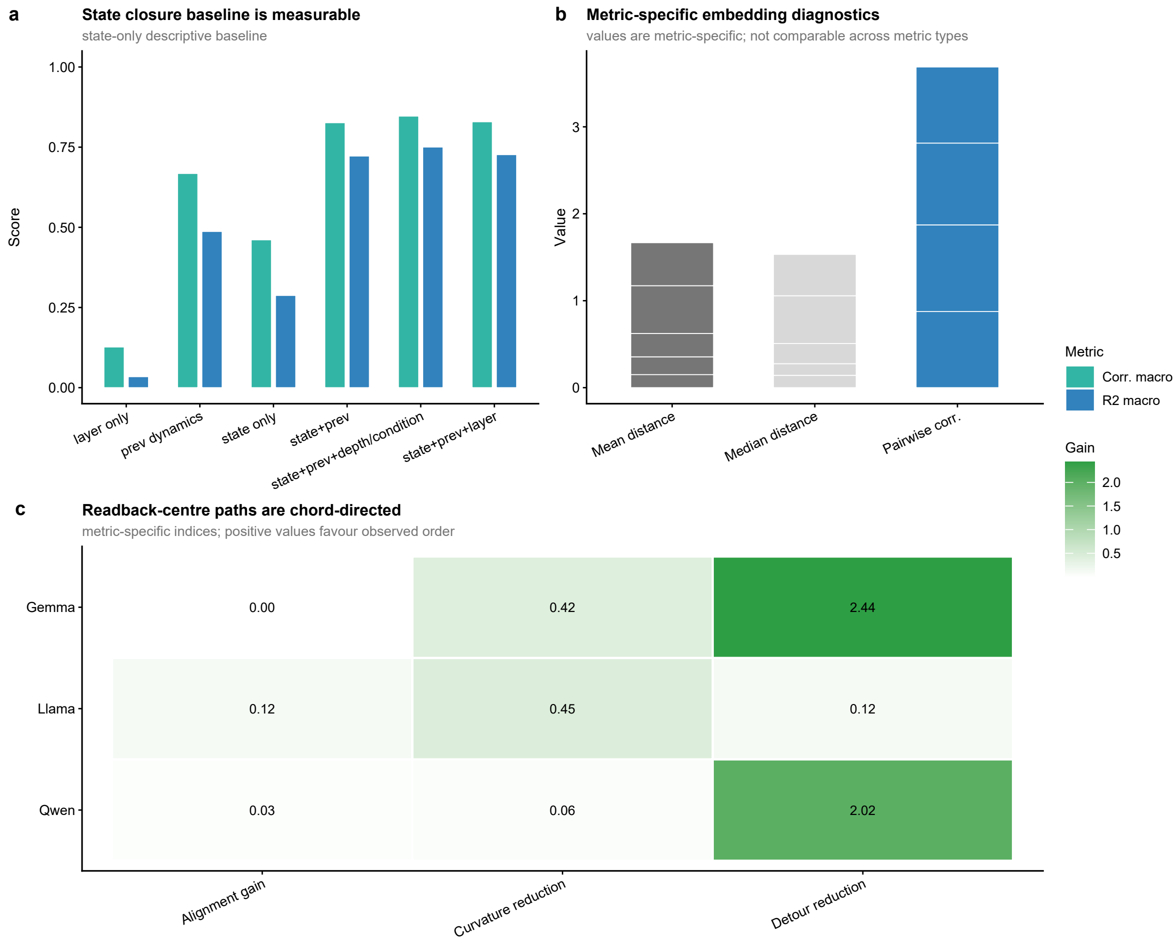
Extended Data



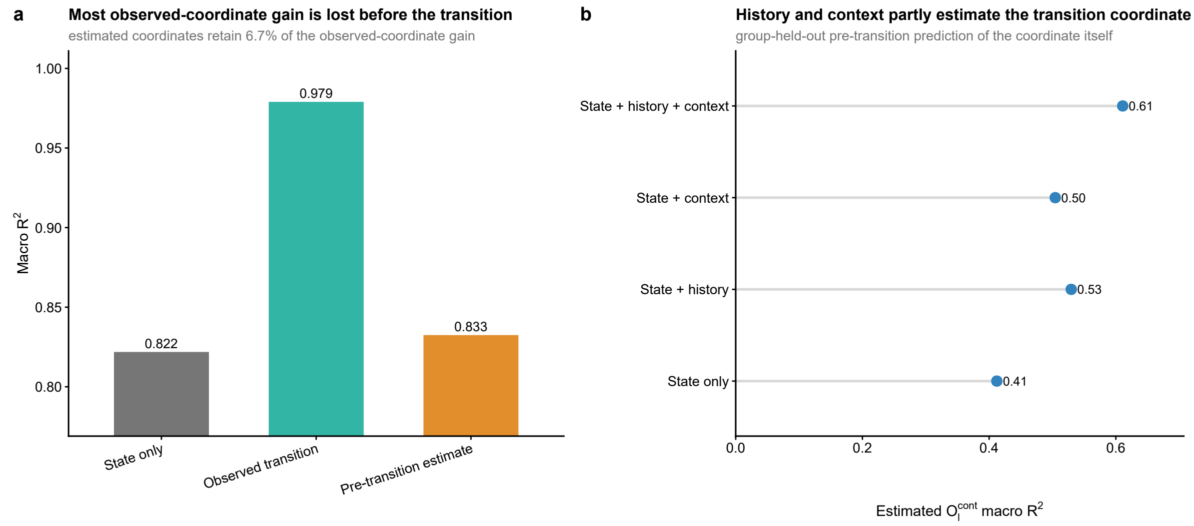
Extended Data Fig. 1 | Low-dimensional orientation view of high-dimensional competition. Three-dimensional surface rendering of a residual-PCA projection, retained only to help readers visualize candidate-path competition, layer order and the separation summarized by ΔU. The surface is a low-dimensional visual abstraction of a high-dimensional process, not a literal state-space manifold or independent mechanism evidence. The layerwise competition readout below the surface is a diagnostic projection. See Source Data Extended Data Fig. 1 and Source Data manifest.



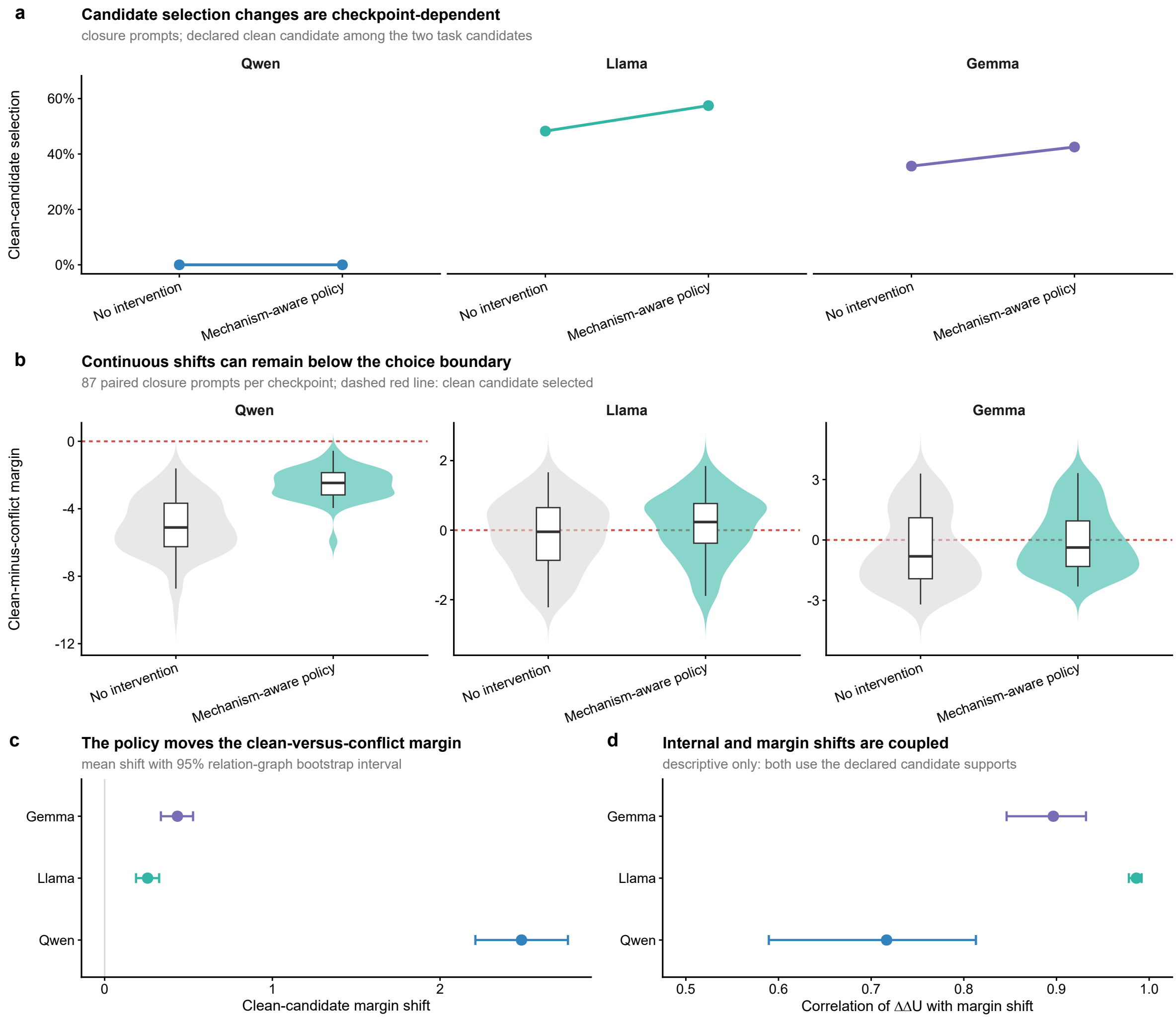
Extended Data Fig. 2 | Coordinate-construction and cross-checkpoint boundaries. a, Construction-aligned decision-window profile, shown separately because it shares its source matrix with ΔU and is not independent coordinate evidence. b, Pairwise direction-isomorphism estimates across checkpoints. Weak and heterogeneous pairwise values, together with the failed prospective recurrence gate in Supplementary Methods Table 2, bound any claim of shared window or coordinate identity. See Source Data Extended Data Fig. 2 and Source Data manifest.



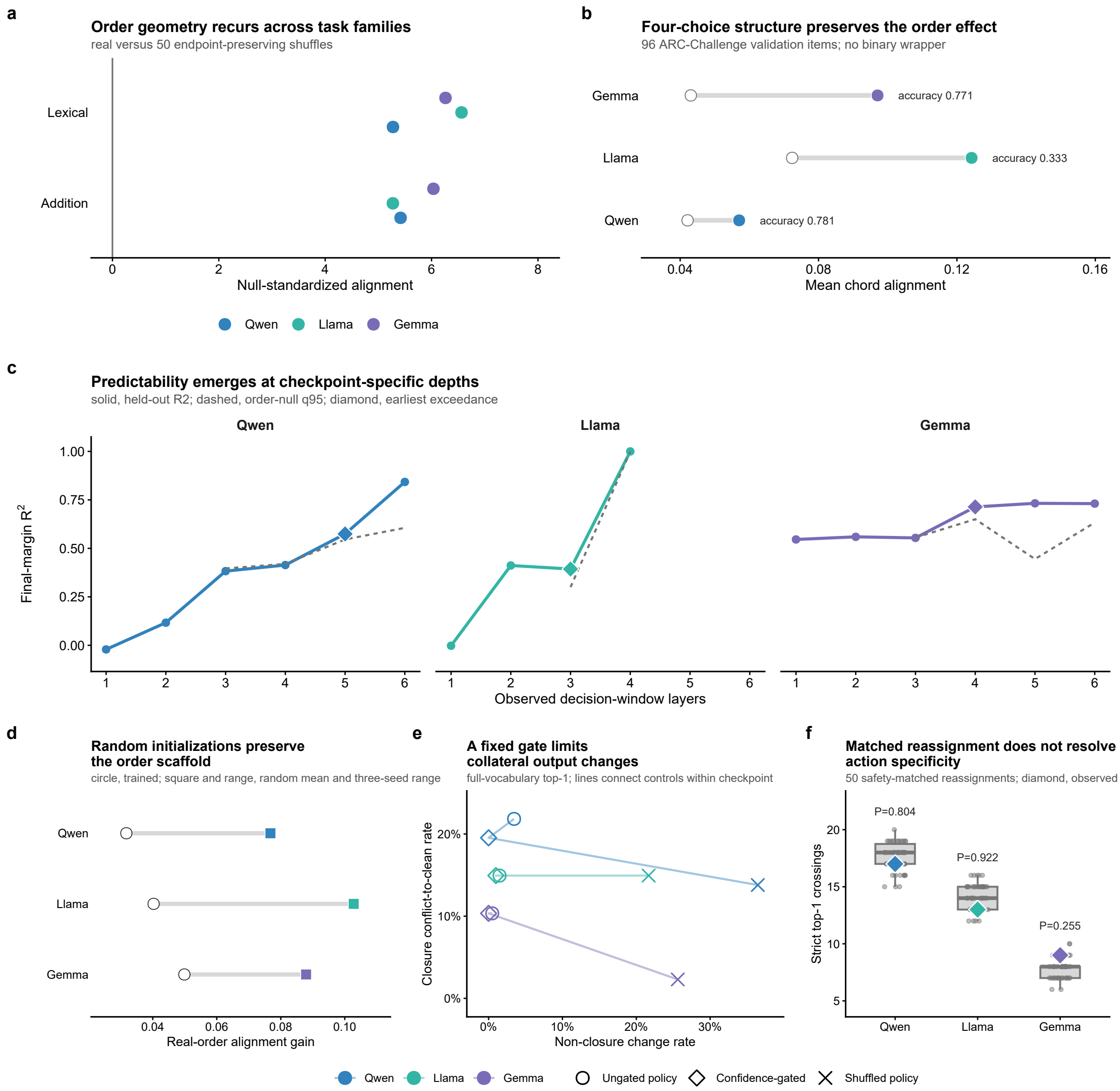
Extended Data Fig. 3 | Metric-specific transition and readback-centre diagnostics. a,b, Embedding and closure diagnostics for realized transition representations. Values are metric-specific and not directly comparable across metric types. c, Chord alignment, detour and turning diagnostics computed on normalized vocabulary-neighbourhood centre trajectories, with fixed endpoints and shuffled intermediate layer order. Panel c is an auxiliary readback-centre analysis, not a direct hidden-state geometry result. See Source Data Extended Data Fig. 3 and Source Data manifest. 910



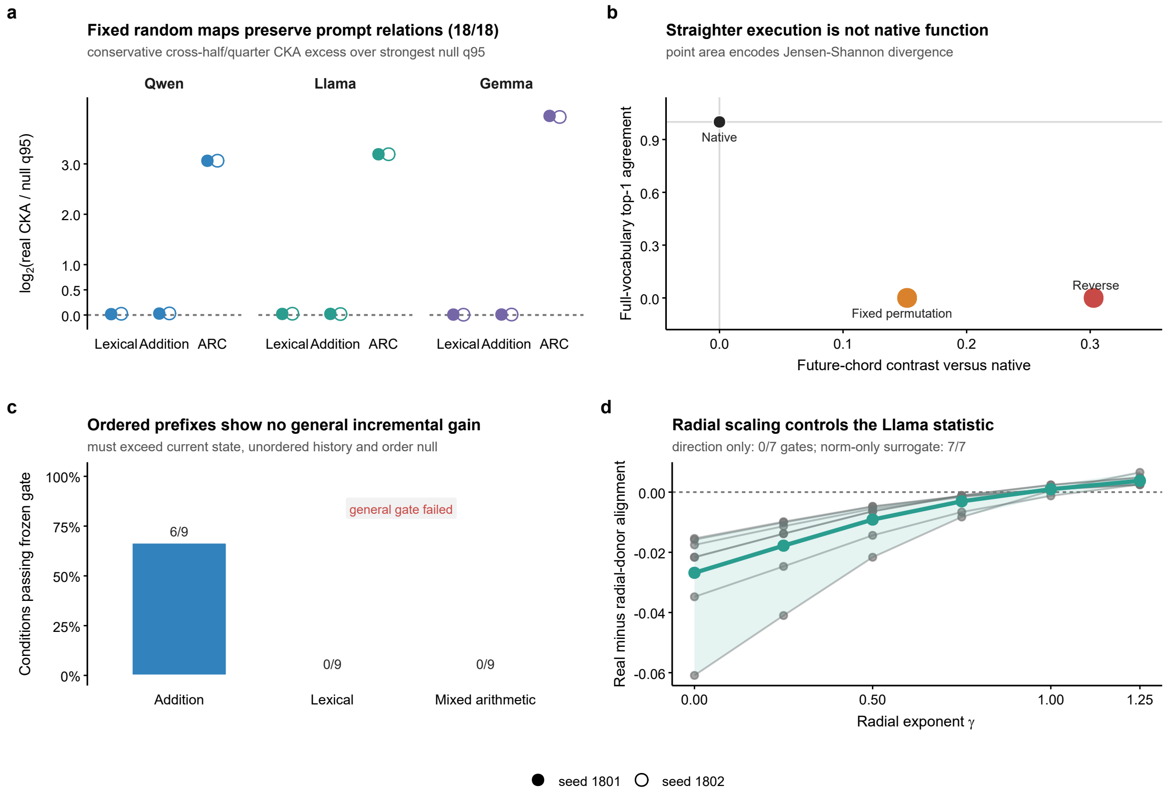
Extended Data Fig. 4 | Predictive boundary of continuous transition coordinates. a, One-step macro R^2 for the state-only predictive baseline, the observed-transition descriptive upper bound and the best bilinear model using a pre-transition estimate of O_l^{cont}. Estimated coordinates retained 25.1% of the observed-coordinate gain; the macro-R^2 increment over state only was 0.0394 (group-bootstrap 95% interval, 0.0368-0.0426). b, Group-held-out prediction of O_l^{cont} from increasingly rich pre-transition feature sets. The target layer and realized transition were excluded from both stages. See Source Data Extended Data Fig. 4 and Source Data manifest.



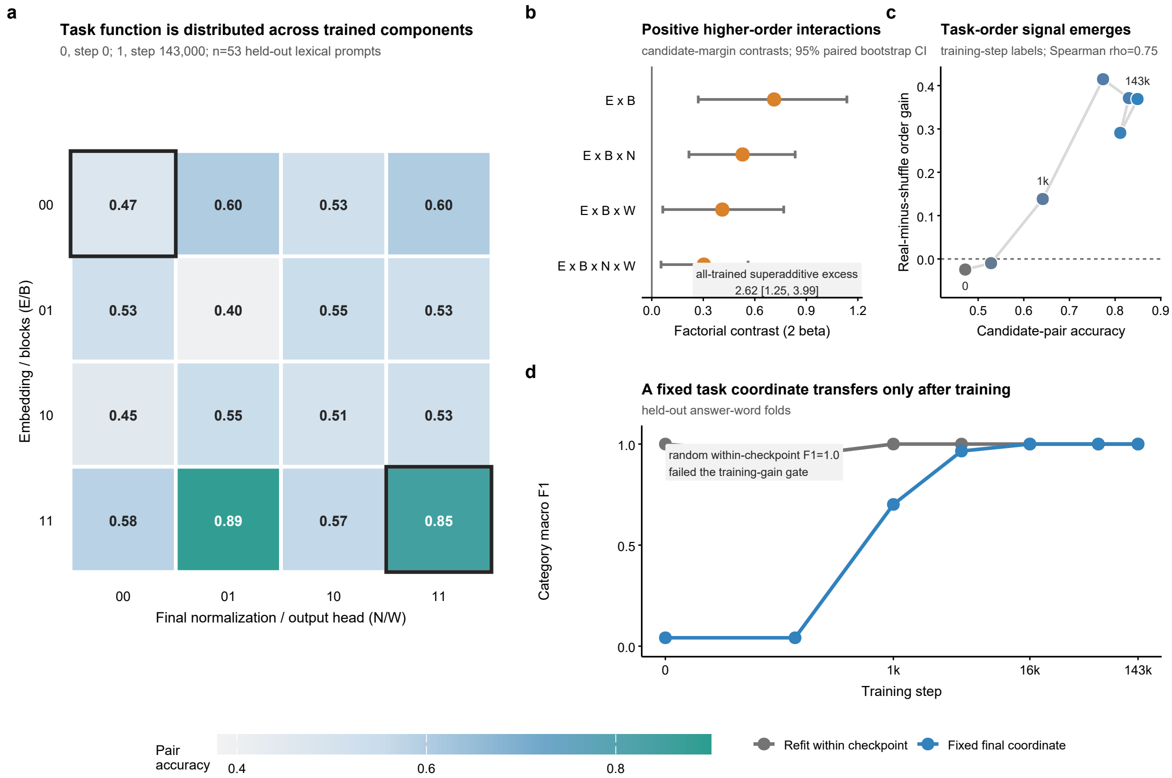
Extended Data Fig. 5 | Candidate-choice boundary under the original action budget. a, Paired clean-candidate selection rates for 87 closure prompts per checkpoint without intervention and under the original mechanism-aware policy. b, Distributions of clean-minus-conflict margins before and after intervention; zero is the candidate-choice boundary. c, Mean policy-induced margin shift with 95% relation-graph bootstrap intervals from 5,000 replicates. d, Correlation between policy-induced ΔU and candidate-margin shifts; this coupling is descriptive because both quantities use the declared candidate supports. All Qwen margins remain below zero under the original budget; the expanded confidence-gated result is in Fig. 5e. See Source Data Extended Data Fig. 5 and Source Data manifest.



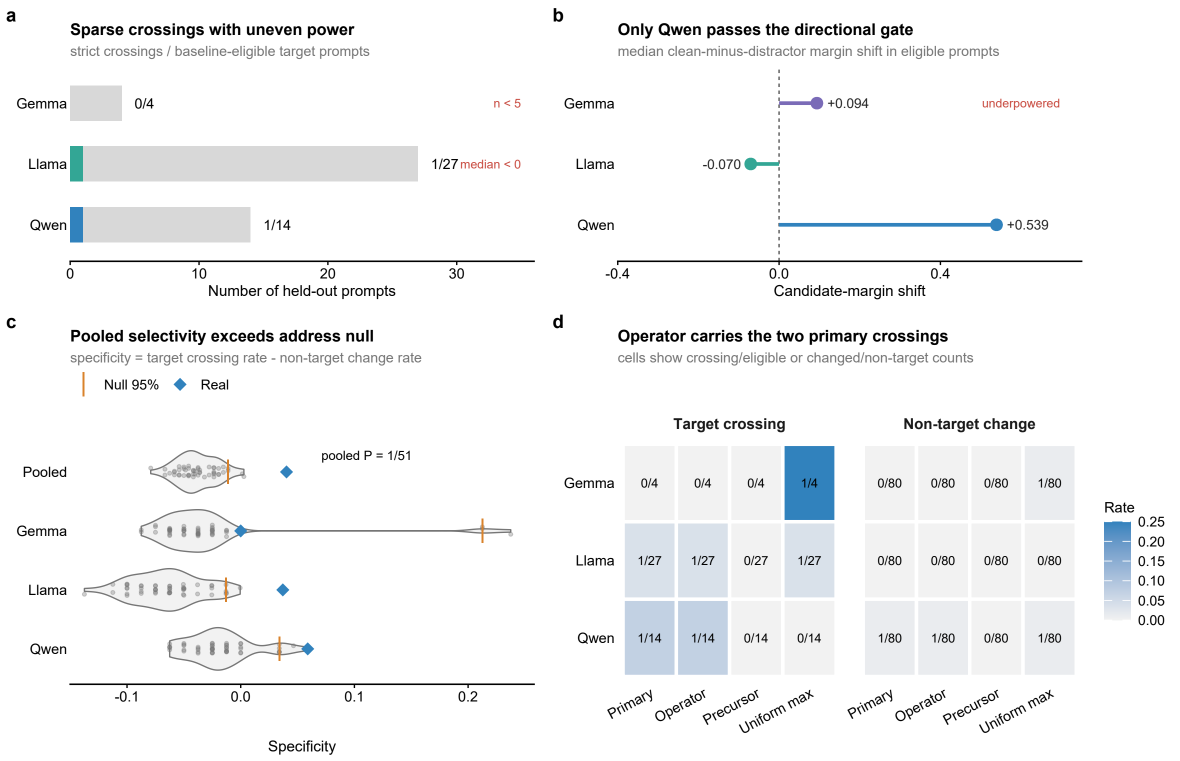
Extended Data Fig. 6 | Task, prediction, initialization and control boundaries. a, Null-standardized chord-alignment effects in controlled lexical and addition tasks. b, Four-choice ARC-Challenge geometry without binary reduction; all real means exceeded 50 endpoint-preserving shuffles. c, Held-out final candidate-margin R^2 across model-specific cutoffs for ordered partial histories and the 95th percentile of 50 independent within-prompt order nulls. d, Alignment gain in trained checkpoints and three random initializations per architecture. e, Ungated, confidence-gated and shuffled-policy full-vocabulary conflict-to-clean and non-closure changes. f, Real strict-flip counts against 50 gate-, condition- and action-multiset-matched reassignments. The numerical relation-graph ΔU axis was not transferred; the forecast is candidate-defined; random-state geometry does not identify learned reasoning; and candidate crossing is not answer correctness. See Source Data Extended Data Fig. 6 and Source Data manifest.



Extended Data Fig. 7 | Falsification separates propagation, order, memory and metric scale. a, Cross-half and quarter-split unbiased CKA for 18 random architecture-task-seed conditions relative to lag-1, block-donor and prompt-fingerprint nulls. b, Reverse and fixed-permutation executions retain positive chord contrast while losing native output fidelity. c, Incremental ordered-prefix gates: 6 of 18 initial conditions passed, all six in addition; lexical prompts passed 0/9 and independent mixed arithmetic passed 0/9. d, Llama direction-only, norm-only and radial-dose controls. These panels reject semantic, functional and predictive interpretations of raw geometric regularity while preserving an architecture-compatible propagation scaffold. See Source Data Extended Data Fig. 7 and Source Data manifest.



Extended Data Fig. 8 | Training organizes compatible task coordinates. a, Candidate-pair accuracy for the complete 2^4 Pythia component swap across input embeddings E, blocks B, final normalization N and output head W. b, Candidate-margin interaction contrasts and the superadditive all-trained excess. c, Model-native candidate-coordinate order gain and pair accuracy across seven training checkpoints. d, Held-out category macro F1 for within-checkpoint coordinates and a fixed final-checkpoint coordinate transferred backwards through training. Random within-checkpoint decodability is a negative control; component swaps are perturbational hybrids; neither establishes a universal semantic metric. See Source Data Extended Data Fig. 8 and Source Data manifest.



Extended Data Fig. 9 | Frozen-protocol actuator transfer is selective but not confirmatory in mixed arithmetic. a, Baseline-eligible target prompts and strict full-vocabulary distractor-to-arithmetic-consistent crossings under the primary confidence-gated actuator; Gemma is underpowered by the pre-declared $n < 5$ rule. b, Median clean-minus-distractor margin shift among eligible prompts; only Qwen combines a powered denominator, positive median shift and a strict crossing. c, Primary specificity against 50 action-address-reassignment nulls per checkpoint and the pooled null; the pooled value exceeds the null 95th percentile (one-sided empirical $P = 1/51$). d, Strict crossings and non-target top-1 changes under the primary, operator-only, precursor-only and uniform-maximum controls. The task-specific arithmetic coordinate was fitted on training problems and is not the relation-graph ΔU axis. Results support partial protocol-level leverage, not confirmed cross-task output control or open-ended correctness. See Source Data Extended Data Fig. 9 and Source Data manifest.